

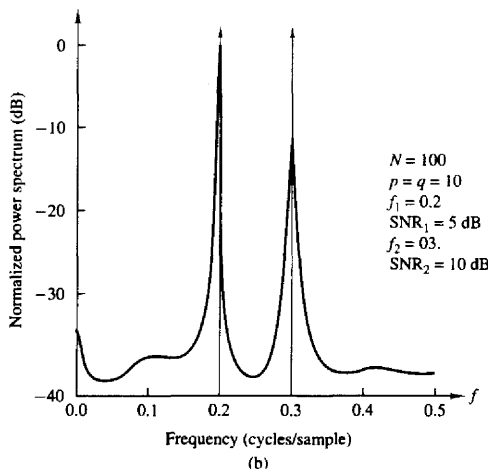
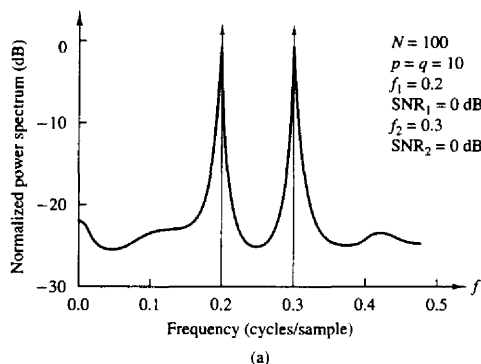


Figure 12.18 ARMA (10, 10) power spectrum estimates from paper by Kay (1980). Reprinted with permission from the IEEE.

the method provides a minimum variance unbiased estimate of the spectral components in the signal.

Following the development of Lacoss, let us consider an FIR filter with coefficients a_k , $0 \leq k \leq p$, to be determined. Unlike the linear prediction problem, we do not constrain a_0 to be unity. Then, if the observed data $x(n)$, $0 \leq n \leq N-1$, are passed through the filter, the response is

$$y(n) = \sum_{k=0}^p a_k x(n-k) \equiv \mathbf{X}'(n) \mathbf{a} \quad (12.4.1)$$

$\mathbf{X}'(n) = [x(n) \ x(n-1) \ \dots \ x(n-p)]$ is the data vector and $\mathbf{a}$ is the filter coefficient vector. If we assume that $E[x(n)] = 0$, the variance of the output sequence is

$$\begin{aligned}\sigma_y^2 &= E[|y(n)|^2] = E[\mathbf{a}^* \mathbf{X}^*(n) \mathbf{X}'(n) \mathbf{a}] \\ &= \mathbf{a}^* \Gamma_{xx} \mathbf{a}\end{aligned}\quad (12.4.2)$$

where Γ_{xx} is the autocorrelation matrix of the sequence $x(n)$, with elements $\gamma_{xx}(m)$.

The filter coefficients are selected so that at the frequency f_i , the frequency response of the FIR filter is normalized to unity, that is,

$$\sum_{k=0}^p a_k e^{-j2\pi k f_i} = 1$$

This constraint can also be written in matrix form as

$$\mathbf{E}'(f_i) \mathbf{a} = 1 \quad (12.4.3)$$

where

$$\mathbf{E}'(f_i) = [1 \ e^{j2\pi f_i} \ \dots \ e^{j2\pi p f_i}]$$

By minimizing the variance σ_y^2 subject to the constraint (12.4.3), we obtain an FIR filter that passes the frequency component f_i undistorted, while components distant from f_i are severely attenuated. The result of this minimization is shown by Lacoss to lead to the coefficient vector

$$\hat{\mathbf{a}} = \Gamma_{xx}^{-1} \mathbf{E}^*(f_i) / \mathbf{E}'(f_i) \Gamma_{xx}^{-1} \mathbf{E}^*(f_i) \quad (12.4.4)$$

If $\hat{\mathbf{a}}$ is substituted into (12.4.2), we obtain the minimum variance

$$\sigma_{\min}^2 = \frac{1}{\mathbf{E}'(f_i) \Gamma_{xx}^{-1} \mathbf{E}^*(f_i)} \quad (12.4.5)$$

The expression in (12.4.5) is the minimum variance power spectrum estimate at the frequency f_i . By changing f_i over the range $0 \leq f_i \leq 0.5$, we can obtain the power spectrum estimate. It should be noted that although $\mathbf{E}(f)$ changes with the choice of frequency, Γ_{xx}^{-1} is computed only once. As demonstrated by Lacoss (1971), the computation of the quadratic form $\mathbf{E}'(f) \Gamma_{xx}^{-1} \mathbf{E}^*(f)$ can be done with a single DFT.

With an estimate $\mathbf{R}_{xx}$ of the autocorrelation matrix substituted in place of Γ_{xx} , we obtain the minimum variance power spectrum estimate of Capon as

$$P_{xx}^{\text{MV}}(f) = \frac{1}{\mathbf{E}'(f) \mathbf{R}_{xx}^{-1} \mathbf{E}^*(f)} \quad (12.4.6)$$

It has been shown by Lacoss (1971) that this power spectrum estimator yields estimates of the spectral peaks proportional to the power at that frequency. In contrast, the AR methods described in Section 12.3 result in estimates of the spectral peaks proportional to the square of the power at that frequency.

This minimum variance method is basically a filter bank implementation for the spectrum estimator. It differs basically from the filter bank interpretation of the periodogram in that the filter coefficients in the Capon method are optimized.

Experiments on the performance of this method compared with the performance of the Burg method have been done by Lacoss (1971) and others. In general, the minimum variance estimate in (12.4.6) outperforms the nonparametric spectral estimators in frequency resolution, but it does not provide the high frequency resolution obtained from the AR methods of Burg and the unconstrained least squares. Extensive comparisons between the Burg method and the minimum variance method have been made in the paper by Lacoss. Furthermore, Burg (1972) demonstrated that for a known correlation sequence, the minimum variance spectrum is related to the AR model spectrum through the equation

$$\frac{1}{\Gamma_{xx}^{MV}(f)} = \frac{1}{p} \sum_{k=0}^p \frac{1}{\Gamma_{xx}^{AR}(f, k)} \quad (12.4.7)$$

where $\Gamma_{xx}^{AR}(f, k)$ is the AR power spectrum obtained with an AR(k) model. Thus the reciprocal of the minimum variance estimate is equal to the average of the reciprocals of all spectra obtained with AR(k) models for $1 \leq k \leq p$. Since low-order AR models, in general, do not provide good resolution, the averaging operation in (12.4.7) reduces the frequency resolution in the spectral estimate. Hence we conclude that the AR power spectrum estimate of order p is superior to the minimum variance estimate of order $p + 1$.

The relationship given by (12.4.7) represents a frequency-domain relationship between the Capon minimum variance estimate and the Burg AR estimate. A time-domain relationship between these two estimates also can be established as shown by Musicus (1985). This has led to a computationally efficient algorithm for the minimum variance estimate.

Additional references to the method of Capon and comparisons with other estimators can be found in the literature. We cite the papers of Capon and Goodman (1971), Marzetta (1983), Marzetta and Lang (1983, 1984), Capon (1983), and McDonough (1983).

12.5 EIGENANALYSIS ALGORITHMS FOR SPECTRUM ESTIMATION

In Section 12.3.8 we demonstrated that an AR(p) process corrupted by additive (white) noise is equivalent to an ARMA(p, p) process. In this section we consider the special case in which the signal components are sinusoids corrupted by additive white noise. The algorithms are based on an eigen-decomposition of the correlation matrix of the noise-corrupted signal.

From our previous discussion on the generation of sinusoids in Chapter 4, we recall that a real sinusoidal signal can be generated via the difference equation,

$$x(n) = -a_1 x(n-1) - a_2 x(n-2) \quad (12.5.1)$$

where $a_1 = 2 \cos 2\pi f_k$, $a_2 = 1$, and initially, $x(-1) = -1$, $x(-2) = 0$. This system has a pair of complex-conjugate poles (at $f = f$ and $f = -f_k$) and therefore generates the sinusoid $x(n) = \cos 2\pi f_k n$, for $n \geq 0$.

In general, a signal consisting of p sinusoidal components satisfies the difference equation

$$x(n) = - \sum_{m=1}^{2p} a_m x(n-m) \quad (12.5.2)$$

and corresponds to the system with system function

$$H(z) = \frac{1}{1 + \sum_{m=1}^{2p} a_m z^{-m}} \quad (12.5.3)$$

The polynomial

$$A(z) = 1 + \sum_{m=1}^{2p} a_m z^{-m} \quad (12.5.4)$$

has $2p$ roots on the unit circle which correspond to the frequencies of the sinusoids.

Now, suppose that the sinusoids are corrupted by a white noise sequence $w(n)$ with $E[|w(n)|^2] = \sigma_w^2$. Then we observe that

$$y(n) = x(n) + w(n) \quad (12.5.5)$$

If we substitute $x(n) = y(n) - w(n)$ in (12.5.2), we obtain

$$y(n) - w(n) = - \sum_{m=1}^{2p} [y(n-m) - w(n-m)] a_m$$

or, equivalently,

$$\sum_{m=0}^{2p} a_m y(n-m) = \sum_{m=0}^{2p} a_m w(n-m) \quad (12.5.6)$$

where, by definition, $a_0 = 1$.

We observe that (12.5.6) is the difference equation for an ARMA(p, p) process in which both the AR and MA parameters are identical. This symmetry is a characteristic of the sinusoidal signals in white noise. The difference equation in (12.5.6) may be expressed in matrix form as

$$\mathbf{Y}' \mathbf{a} = \mathbf{W}' \mathbf{a} \quad (12.5.7)$$

where $\mathbf{Y}' = [y(n) \ y(n-1) \ \cdots \ y(n-2p)]$ is the observed data vector of dimension $(2p+1)$, $\mathbf{W}' = [w(n) \ w(n-1) \ \cdots \ w(n-2p)]$ is the noise vector, and $\mathbf{a} = [1 \ a_1 \ \cdots \ a_{2p}]$ is the coefficient vector.

If we premultiply (12.5.7) by $\mathbf{Y}$ and take the expected value, we obtain

$$E(\mathbf{Y}\mathbf{Y}')\mathbf{a} = E(\mathbf{Y}\mathbf{W}')\mathbf{a} = E[(\mathbf{X} + \mathbf{W})\mathbf{W}']\mathbf{a} \quad (12.5.8)$$

$$\Gamma_{yy}\mathbf{a} = \sigma_w^2 \mathbf{a}$$

where we have used the assumption that the sequence $w(n)$ is zero mean and white, and $\mathbf{X}$ is a deterministic signal.

The equation in (12.5.8) is in the form of an eigenequation, that is,

$$(\Gamma_{yy} - \sigma_w^2 \mathbf{I})\mathbf{a} = \mathbf{0} \quad (12.5.9)$$

where σ_w^2 is an eigenvalue of the autocorrelation matrix Γ_{yy} . Then the parameter vector $\mathbf{a}$ is an eigenvector associated with the eigenvalue σ_w^2 . The eigenequation in (12.5.9) forms the basis for the Pisarenko harmonic decomposition method.

12.5.1 Pisarenko Harmonic Decomposition Method

For p randomly-phased sinusoids in additive white noise, the autocorrelation values are

$$\begin{aligned} \gamma_{yy}(0) &= \sigma_w^2 + \sum_{i=1}^p P_i \\ \gamma_{yy}(k) &= \sum_{i=1}^p P_i \cos 2\pi f_i k \quad k \neq 0 \end{aligned} \quad (12.5.10)$$

where $P_i = A_i^2/2$ is the average power in the i th sinusoid and A_i is the corresponding amplitude. Hence we may write

$$\begin{bmatrix} \cos 2\pi f_1 & \cos 2\pi f_2 & \cdots & \cos 2\pi f_p \\ \cos 4\pi f_1 & \cos 4\pi f_2 & \cdots & \cos 4\pi f_p \\ \vdots & \vdots & & \vdots \\ \cos 2\pi p f_1 & \cos 2\pi p f_2 & \cdots & \cos 2\pi p f_p \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ \vdots \\ P_p \end{bmatrix} = \begin{bmatrix} \gamma_{yy}(1) \\ \gamma_{yy}(2) \\ \vdots \\ \gamma_{yy}(p) \end{bmatrix} \quad (12.5.11)$$

If we know the frequencies f_i , $1 \leq i \leq p$, we can use this equation to determine the powers of the sinusoids. In place of $\gamma_{xx}(m)$, we use the estimates $r_{xx}(m)$. Once the powers are known, the noise variance can be obtained from (12.5.10) as

$$\sigma_w^2 = r_{yy}(0) - \sum_{i=1}^p P_i \quad (12.5.12)$$

The problem that remains is to determine the p frequencies f_i , $1 \leq i \leq p$, which, in turn, require knowledge of the eigenvector $\mathbf{a}$ corresponding to the eigenvalue σ_w^2 . Pisarenko (1973) observed [see also Papoulis (1984) and Grenander and Szegő (1958)] that for an ARMA process consisting of p sinusoids in additive white noise, the variance σ_w^2 corresponds to the minimum eigenvalue of Γ_{yy} when the dimension of the autocorrelation matrix equals or exceeds $(2p+1) \times (2p+1)$. The desired ARMA coefficient vector corresponds to the eigenvector associated with the minimum eigenvalue. Therefore, the frequencies f_i , $1 \leq i \leq p$ are obtained from the roots of the polynomial in (12.5.4), where the coefficients are the elements of the eigenvector $\mathbf{a}$ corresponding to the minimum eigenvalue σ_w^2 .

In summary, the Pisarenko harmonic decomposition method proceeds as follows. First we estimate Γ_{yy} from the data (i.e., we form the autocorrelation matrix $\mathbf{R}_{yy}$). Then we find the minimum eigenvalue and the corresponding minimum eigenvector. The minimum eigenvector yields the parameters of the

ARMA(2p, 2p) model. From (12.5.4) we can compute the roots that constitute the frequencies $\{f_i\}$. By using these frequencies, we can solve (12.5.11) for the signal powers $\{P_i\}$ by substituting the estimates $r_{yy}(m)$ for $\gamma_{yy}(m)$.

As will be seen in the following example, the Pisarenko method is based on the use of a noise subspace eigenvector to estimate the frequencies of the sinusoids.

Example 12.5.1

Suppose that we are given the autocorrelation values $\gamma_{yy}(0) = 3$, $\gamma_{yy}(1) = 1$, and $\gamma_{yy}(2) = 0$ for a process consisting of a single sinusoid in additive white noise. Determine the frequency, its power, and the variance of the additive noise.

Solution The correlation matrix is

$$\Gamma_{yy} = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

The minimum eigenvalue is the smallest root of the characteristic polynomial

$$g(\lambda) = \begin{vmatrix} 3-\lambda & 1 & 0 \\ 1 & 3-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{vmatrix} = (3-\lambda)(\lambda^2 - 6\lambda + 7) = 0$$

Therefore, the eigenvalues are $\lambda_1 = 3$, $\lambda_2 = 3 + \sqrt{2}$, $\lambda_3 = 3 - \sqrt{2}$.

The variance of the noise is

$$\sigma_v^2 = \lambda_{\min} = 3 - \sqrt{2}$$

The corresponding eigenvector is the vector that satisfies (12.5.9), that is,

$$\begin{bmatrix} \sqrt{2} & 1 & 0 \\ 1 & \sqrt{2} & 1 \\ 0 & 1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The solution is $a_1 = -\sqrt{2}$ and $a_2 = 1$.

The next step is to use the value a_1 and a_2 to determine the roots of the polynomial in (12.5.4). We have

$$z^2 - \sqrt{2}z + 1 = 0$$

Thus

$$z_1, z_2 = \frac{1}{\sqrt{2}} \pm j \frac{1}{\sqrt{2}}$$

Note that $|z_1| = |z_2| = 1$, so that the roots are on the unit circle. The corresponding frequency is obtained from

$$z_i = e^{j2\pi f_i} = \frac{1}{\sqrt{2}} + j \frac{1}{\sqrt{2}}$$

which yields $f_1 = \frac{1}{8}$. Finally, the power of the sinusoid is

$$P_1 \cos 2\pi f_1 = \gamma_{yy}(1) = 1$$

$$P_1 = \sqrt{2}$$

and its amplitude is $A = \sqrt{2P_1} = \sqrt{2\sqrt{2}}$.

As a check on our computations, we have

$$\begin{aligned}\sigma_w^2 &= \gamma_{yy}(0) - P_1 \\ &= 3 - \sqrt{2}\end{aligned}$$

which agrees with $\lambda_{\min}$.

12.5.2 Eigen-decomposition of the Autocorrelation Matrix for Sinusoids in White Noise

In the previous discussion we assumed that the sinusoidal signal consists of p real sinusoids. For mathematical convenience we shall now assume that the signal consists of p complex sinusoids of the form

$$x(n) = \sum_{i=1}^p A_i e^{j(2\pi f_i n + \phi_i)} \quad (12.5.13)$$

where the amplitudes $\{A_i\}$ and the frequencies $\{f_i\}$ are unknown and the phases $\{\phi_i\}$ are statistically independent random variables uniformly distributed on $(0, 2\pi)$. Then the random process $x(n)$ is wide-sense stationary with autocorrelation function

$$\gamma_{xx}(m) = \sum_{i=1}^p P_i e^{j2\pi f_i m} \quad (12.5.14)$$

where, for complex sinusoids, $P_i = A_i^2$ is the power of the i th sinusoid.

Since the sequence observed is $y(n) = x(n) + w(n)$, where $w(n)$ is a white noise sequence with spectral density σ_w^2 , the autocorrelation function for $y(n)$ is

$$\gamma_{yy}(m) = \gamma_{xx}(m) + \sigma_w^2 \delta(m) \quad m = 0, \pm 1, \dots, \pm(M-1) \quad (12.5.15)$$

Hence the $M \times M$ autocorrelation matrix for $y(n)$ can be expressed as

$$\Gamma_{yy} = \Gamma_{xx} + \sigma_w^2 \mathbf{I} \quad (12.5.16)$$

where Γ_{xx} is the autocorrelation matrix for the signal $x(n)$ and $\sigma_w^2 \mathbf{I}$ is the autocorrelation matrix for the noise. Note that if select $M > p$, Γ_{xx} which is of dimension $M \times M$ is not of full rank, because its rank is p . However, Γ_{yy} is full rank because $\sigma_w^2 \mathbf{I}$ is of rank M .

In fact, the signal matrix Γ_{xx} can be represented as

$$\Gamma_{xx} = \sum_{i=1}^p P_i \mathbf{s}_i \mathbf{s}_i^H \quad (12.5.17)$$

where H denotes the conjugate transpose and $\mathbf{s}_i$ is a signal vector of dimension M defined as

$$\mathbf{s}_i = [1, e^{j2\pi f_i}, e^{j4\pi f_i}, \dots, e^{j2\pi(M-1)f_i}] \quad (12.5.18)$$

Since each vector (outer product) $\mathbf{s}_i \mathbf{s}_i^H$ is a matrix of rank 1 and since there are p vector products, the matrix Γ_{xx} is of rank p . Note that if the sinusoids were real, the correlation matrix Γ_{xx} has rank $2p$.

Now, let us perform an eigen-decomposition of the matrix Γ_{yy} . Let the eigenvalues $\{\lambda_i\}$ be ordered in decreasing value with $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \dots \geq \lambda_M$ and let the corresponding eigenvectors be denoted as $\{\mathbf{v}_i, i = 1, \dots, M\}$. We assume that the eigenvectors are normalized so that $\mathbf{v}_i^H \cdot \mathbf{v}_j = \delta_{ij}$. In the absence of noise the eigenvalues $\lambda_i, i = 1, 2, \dots, p$, are nonzero while $\lambda_{p+1} = \lambda_{p+2} = \dots = \lambda_M = 0$. Furthermore, it follows that the signal correlation matrix can be expressed as

$$\Gamma_{xx} = \sum_{i=1}^p \lambda_i \mathbf{v}_i \mathbf{v}_i^H \quad (12.5.19)$$

Thus, the eigenvectors $\mathbf{v}_i, i = 1, 2, \dots, p$ span the signal subspace as do the signal vectors $\mathbf{s}_i, i = 1, 2, \dots, p$. These p eigenvectors for the signal subspace are called the *principal eigenvectors* and the corresponding eigenvalues are called the *principal eigenvalues*.

In the presence of noise, the noise autocorrelation matrix in (12.5.16) can be represented as

$$\sigma_w^2 \mathbf{I} = \sigma_w^2 \sum_{i=1}^M \mathbf{v}_i \mathbf{v}_i^H \quad (12.5.20)$$

By substituting (12.5.19) and (12.5.20) into (12.5.16), we obtain

$$\begin{aligned} \Gamma_{yy} &= \sum_{i=1}^p \lambda_i \mathbf{v}_i \mathbf{v}_i^H + \sum_{i=1}^M \sigma_w^2 \mathbf{v}_i \mathbf{v}_i^H \\ &= \sum_{i=1}^p (\lambda_i + \sigma_w^2) \mathbf{v}_i \mathbf{v}_i^H + \sum_{i=p+1}^M \sigma_w^2 \mathbf{v}_i \mathbf{v}_i^H \end{aligned} \quad (12.5.21)$$

This eigen-decomposition separates the eigenvectors into two sets. The set $\{\mathbf{v}_i, i = 1, 2, \dots, p\}$, which are the principal eigenvectors, span the signal subspace, while the set $\{\mathbf{v}_i, i = p+1, \dots, M\}$, which are orthogonal to the principal eigenvectors, are said to belong to the noise subspace. Since the signal vectors $\{\mathbf{s}_i, i = 1, 2, \dots, p\}$ are in the signal subspace, it follows that the $\{\mathbf{s}_i\}$ are simply linear combinations of the principal eigenvectors and are also orthogonal to the vectors in the noise subspace.

In this context we see that the Pisarenko method is based on an estimation of the frequencies by using the orthogonality property between the signal vectors and the vectors in the noise subspace. For complex sinusoids, if we select $M = p + 1$ (for real sinusoids we select $M = 2p + 1$), there is only a single eigenvector in the noise subspace (corresponding to the minimum eigenvalue) which must be orthogonal to the signal vectors. Thus we have

$$\mathbf{s}_i^H \mathbf{v}_{p+1} = \sum_{k=0}^p \mathbf{v}_{p+1}(k+1) e^{-j2\pi f_i k} = 0 \quad i = 1, 2, \dots, p \quad (12.5.22)$$

But (12.5.22) implies that the frequencies $\{f_i\}$ can be determined by solving for

the zeros of the polynomial

$$V(z) = \sum_{n=0}^p v_{p+1}(k+1)z^{-k} \quad (12.5.23)$$

all of which lie on the unit circle. The angles of these roots are $2\pi f_i$, $i = 1, 2, \dots, p$.

When the number of sinusoids is unknown, the determination of p may prove to be difficult, especially if the signal level is not much higher than the noise level. In theory, if $M > p + 1$, there is a multiplicity $(M - p)$ of the minimum eigenvalue. However, in practice the $(M - p)$ small eigenvalues of $\mathbf{R}_{yy}$ will probably be different. By computing all the eigenvalues it may be possible to determine p by grouping the $M - p$ small (noise) eigenvalues into a set and averaging them to obtain an estimate of σ_w^2 . Then, the average value can be used in (12.5.9) along with $\mathbf{R}_{yy}$ to determine the corresponding eigenvector.

12.5.3 MUSIC Algorithm

The multiple signal classification (MUSIC) method is also a noise subspace frequency estimator. To develop the method, let us first consider the “weighted” spectral estimate

$$P(f) = \sum_{k=p+1}^M w_k |\mathbf{s}^H(f) \mathbf{v}_k|^2 \quad (12.5.24)$$

where $\{\mathbf{v}_k, k = p + 1, \dots, M\}$ are the eigenvectors in the noise subspace, $\{w_k\}$ are a set of positive weights, and $\mathbf{s}(f)$ is the complex sinusoidal vector

$$\mathbf{s}(f) = [1, e^{j2\pi f}, e^{j4\pi f}, \dots, e^{j2\pi(M-1)f}] \quad (12.5.25)$$

Note that at $f = f_i$, $\mathbf{s}(f_i) \equiv \mathbf{s}_i$, so that at any one of the p sinusoidal frequency components of the signal, we have

$$P(f_i) = 0 \quad i = 1, 2, \dots, p \quad (12.5.26)$$

Hence, the reciprocal of $P(f)$ is a sharply peaked function of frequency and provides a method for estimating the frequencies of the sinusoidal components. Thus

$$\frac{1}{P(f)} = \frac{1}{\sum_{k=p+1}^M w_k |\mathbf{s}^H(f) \mathbf{v}_k|^2} \quad (12.5.27)$$

Although theoretically $1/P(f)$ is infinite at $f = f_i$, in practice the estimation errors result in finite values for $1/P(f)$ at all frequencies.

The MUSIC sinusoidal frequency estimator proposed by Schmidt (1981, 1986) is a special case of (12.5.27) in which the weights $w_k = 1$ for all k . Hence

$$P_{\text{MUSIC}}(f) = \frac{1}{\sum_{k=p+1}^M |\mathbf{s}^H(f) \mathbf{v}_k|^2} \quad (12.5.28)$$

The estimate of the sinusoidal frequencies are the peaks of $P_{\text{MUSIC}}(f)$. Once the sinusoidal frequencies are estimated, the power of each of the sinusoids can be obtained by solving (12.5.11).

12.5.4 ESPRIT Algorithm

ESPRIT (estimation of signal parameters via rotational invariance techniques) is yet another method for estimating frequencies of a sum of sinusoids by use of an eigen-decomposition approach. As we observe from the development that follows, which is due to Roy et al. (1986), ESPRIT exploits an underlying rotational invariance of signal subspaces spanned by two temporally displaced data vectors.

We again consider the estimation of p complex-valued sinusoids in additive white noise. The received sequence is given by the vector

$$\begin{aligned}\mathbf{y}(n) &= [y(n), y(n+1), \dots, y(n+M-1)]^T \\ &= \mathbf{x}(n) + \mathbf{w}(n)\end{aligned}\quad (12.5.29)$$

where $\mathbf{x}(n)$ is the signal vector and $\mathbf{w}(n)$ is the noise vector. To exploit the deterministic character of the sinusoids, we define the time-displaced vector $\mathbf{z}(n) = \mathbf{y}(n+1)$. Thus

$$\begin{aligned}\mathbf{z}(n) &= [z(n), z(n+1), \dots, z(n+M-1)]^T \\ &= [y(n+1), y(n+2), \dots, y(n+M)]^T\end{aligned}\quad (12.5.30)$$

With these definitions we can express the vectors $\mathbf{y}(n)$ and $\mathbf{z}(n)$ as

$$\begin{aligned}\mathbf{y}(n) &= \mathbf{S}\mathbf{a} + \mathbf{w}(n) \\ \mathbf{z}(n) &= \mathbf{S}\Phi\mathbf{a} + \mathbf{w}(n)\end{aligned}\quad (12.5.31)$$

where $\mathbf{a} = [a_1, a_2, \dots, a_p]^T$, $a_i = A_i e^{j\phi_i}$, and Φ is a diagonal $p \times p$ matrix consisting of the relative phase between adjacent time samples of each of the complex sinusoids,

$$\Phi = \text{diag}[e^{j2\pi f_1}, e^{j2\pi f_2}, \dots, e^{j2\pi f_p}] \quad (12.5.32)$$

Note that the matrix Φ relates the time-displaced vectors $\mathbf{y}(n)$ and $\mathbf{z}(n)$ and can be called a rotation operator. We also note that Φ is unitary. The matrix $\mathbf{S}$ is the $M \times p$ Vandermonde matrix specified by the column vectors

$$\mathbf{s}_i = [1, e^{j2\pi f_i}, e^{j4\pi f_i}, \dots, e^{j2\pi(M-1)f_i}] \quad i = 1, 2, \dots, p \quad (12.5.33)$$

Now the autocovariance matrix for the data vector $\mathbf{y}(n)$ is

$$\begin{aligned}\Gamma_{yy} &= E[\mathbf{y}(n)\mathbf{y}^H(n)] \\ &= \mathbf{S}\mathbf{P}\mathbf{S}^H + \sigma_w^2\mathbf{I}\end{aligned}\quad (12.5.34)$$

where $\mathbf{P}$ is the $p \times p$ diagonal matrix consisting of the powers of the complex sinusoids,

$$\begin{aligned}\mathbf{P} &= \text{diag}[|a_1|^2, |a_2|^2, \dots, |a_p|^2] \\ &= \text{diag}[P_1, P_2, \dots, P_p]\end{aligned}\quad (12.5.35)$$

We observe that $\mathbf{P}$ is a diagonal matrix since complex sinusoids of different frequencies are orthogonal over the infinite interval. However, we should emphasize that the ESPRIT algorithm does not require $\mathbf{P}$ to be diagonal. Hence the algorithm is applicable to the case in which the covariance matrix is estimated from finite data records.

The crosscovariance matrix of the signal vectors $\mathbf{y}(n)$ and $\mathbf{z}(n)$ is

$$\Gamma_{yz} = E[\mathbf{y}(n)\mathbf{z}^H(n)] = \mathbf{S}\mathbf{P}\mathbf{\Phi}^H\mathbf{S}^H + \Gamma_w \quad (12.5.36)$$

where

$$\begin{aligned}\Gamma_w &= E[\mathbf{w}(n)\mathbf{w}^H(n+1)] \\ &= \sigma_w^2 \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 0 \\ 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 0 \end{bmatrix} \equiv \sigma_w^2 \mathbf{Q}\end{aligned}\quad (12.5.37)$$

The auto and crosscovariance matrices Γ_{yy} and Γ_{yz} are given as

$$\Gamma_{yy} = \begin{bmatrix} \gamma_{yy}(0) & \gamma_{yy}(1) & \dots & \gamma_{yy}(M-1) \\ \gamma_{yy}^*(1) & \gamma_{yy}(0) & \dots & \gamma_{yy}(M-2) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{yy}^*(M-1) & \gamma_{yy}(M-2) & \dots & \gamma_{yy}(0) \end{bmatrix} \quad (12.5.38)$$

$$\Gamma_{yz} = \begin{bmatrix} \gamma_{yy}(1) & \gamma_{yy}(2) & \dots & \gamma_{yy}(M) \\ \gamma_{yy}(0) & \gamma_{yy}(1) & \dots & \gamma_{yy}(M-1) \\ \vdots & \vdots & \ddots & \vdots \\ \gamma_{yy}^*(M-2) & \gamma_{yy}^*(M-3) & \dots & \gamma_{yy}(1) \end{bmatrix} \quad (12.5.39)$$

where $\gamma_{yy}(m) = E[y^*(n)y(n+m)]$. Note that both Γ_{yy} and Γ_{yz} are Toeplitz matrices.

Based on this formulation, the problem is to determine the frequencies $\{f_i\}$ and their powers $\{P_i\}$ from the autocorrelation sequence $\{\gamma_{yy}(m)\}$.

From the underlying model, it is clear that the matrix $\mathbf{S}\mathbf{P}\mathbf{S}^H$ has rank p . Consequently, Γ_{yy} given by (12.5.34) has $(M-p)$ identical eigenvalues equal to σ_w^2 . Hence

$$\Gamma_{yy} - \sigma_w^2 \mathbf{I} = \mathbf{S}\mathbf{P}\mathbf{S}^H \equiv \mathbf{C}_{yy} \quad (12.5.40)$$

From (12.5.36) we also have

$$\Gamma_{yz} - \sigma_w^2 \Gamma_w = \mathbf{S}\mathbf{P}\mathbf{\Phi}^H\mathbf{S}^H \equiv \mathbf{C}_{yz} \quad (12.5.41)$$

Now, let us consider the matrix $\mathbf{C}_{yy} - \lambda \mathbf{C}_{yz}$, which can be written as

$$\mathbf{C}_{yy} - \lambda \mathbf{C}_{yz} = \mathbf{S}(\mathbf{I} - \lambda \mathbf{\Phi}^H)\mathbf{S}^H \quad (12.5.42)$$

Clearly, the column space of $\mathbf{SPS}^H$ is identical to the column space of $\mathbf{SP}\Phi^H\mathbf{S}^H$. Consequently, the rank of $\mathbf{C}_{yy} - \lambda\mathbf{C}_{yz}$ is equal to p . However, we note that if $\lambda = \exp(j2\pi f_i)$, the i th row of $(\mathbf{I} - \lambda\Phi^H)$ is zero and, hence the rank of $[\mathbf{I} - \Phi^H \exp(j2\pi f_i)]$ is $p - 1$. But $\lambda_i = \exp(j2\pi f_i)$, $i = 1, 2, \dots, p$, are the generalized eigenvalues of the matrix pair $(\mathbf{C}_{yy}, \mathbf{C}_{yz})$. Thus the p generalized eigenvalues $\{\lambda_i\}$ that lie on the unit circle correspond to the elements of the rotation operator Φ . The remaining $M - p$ generalized eigenvalues of the pair $\{\mathbf{C}_{yy}, \mathbf{C}_{yz}\}$ which correspond to the common null space of these matrices, are zero [i.e., the $(M - p)$ eigenvalues are at the origin in the complex plane].

Based on these mathematical relationships we can formulate an algorithm (ESPRIT) for estimating the frequencies $\{f_i\}$. The procedure is as follows:

1. From the data, compute the autocorrelation values $r_{yy}(m)$, $m = 1, 2, \dots, M$, and form the matrices $\mathbf{R}_{yy}$ and $\mathbf{R}_{yz}$ corresponding to estimates of Γ_{yy} and Γ_{yz} .
2. Compute the eigenvalues of $\mathbf{R}_{yy}$. For $M > p$, the minimum eigenvalue is an estimate of σ_w^2 .
3. Compute $\hat{\mathbf{C}}_{yy} = \mathbf{R}_{yy} - \hat{\sigma}_w^2\mathbf{I}$ and $\hat{\mathbf{C}}_{yz} = \mathbf{R}_{yz} - \hat{\sigma}_w^2\mathbf{Q}$, where $\mathbf{Q}$ is defined in (12.5.37).
4. Compute the generalized eigenvalues of the matrix pair $(\hat{\mathbf{C}}_{yy}, \hat{\mathbf{C}}_{yz})$. The p generalized eigenvalues of these matrices that lie on (or near) the unit circle determine the (estimate) elements of Φ and hence the sinusoidal frequencies. The remaining $M - p$ eigenvalues will lie at (or near) the origin.

One method for determining the power in the sinusoidal components is to solve the equation in (12.5.11) with $r_{yy}(m)$ substituted for $\gamma_{yy}(m)$.

Another method is based on the computation of the generalized eigenvectors $\{\mathbf{v}_i\}$ corresponding to the generalized eigenvalues $\{\lambda_i\}$. We have

$$(\mathbf{C}_{yy} - \lambda_i \mathbf{C}_{yz})\mathbf{v}_i = \mathbf{SP}(\mathbf{I} - \lambda_i \Phi^H)\mathbf{S}^H \mathbf{v}_i = 0 \quad (12.5.43)$$

Since the column space of $(\mathbf{C}_{yy} - \lambda_i \mathbf{C}_{yz})$ is identical to the column space spanned by the vectors $\{\mathbf{s}_j, j \neq i\}$ given by (12.5.33), it follows that the generalized eigenvector $\mathbf{v}_i$ is orthogonal to $\mathbf{s}_j$, $j \neq i$. Since $\mathbf{P}$ is diagonal, it follows from (12.5.43) that the signal powers are

$$P_i = \frac{\mathbf{v}_i^H \mathbf{C}_{yy} \mathbf{v}_i}{|\mathbf{v}_i^H \mathbf{s}_i|^2} \quad i = 1, 2, \dots, p \quad (12.5.44)$$

12.5.5 Order Selection Criteria

The eigenanalysis methods described in this section for estimating the frequencies and the powers of the sinusoids, also provide information about the number of sinusoidal components. If there are p sinusoids, the eigenvalues associated with the

signal subspace are $\{\lambda_i + \sigma_w^2, i = 1, 2, \dots, p\}$ while the remaining $(M-p)$ eigenvalues are all equal to σ_w^2 . Based on this eigenvalue decomposition, a test can be designed that compares the eigenvalues with a specified threshold. An alternative method also uses the eigenvector decomposition of the estimated autocorrelation matrix of the observed signal and is based on matrix perturbation analysis. This method is described in the paper by Fuchs (1988).

Another approach based on an extension and modification of the AIC criterion to the eigen-decomposition method, has been proposed by Wax and Kailath (1985). If the eigenvalues of the sample autocorrelation matrix are ranked so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_M$, where $M > p$, the number of sinusoids in the signal subspace is estimated by selecting the minimum value of MDL(p), given as

$$\text{MDL}(p) = -\log \left[\frac{G(p)}{A(p)} \right]^N + E(p) \quad (12.5.45)$$

where

$$\begin{aligned} G(p) &= \prod_{i=p+1}^M \lambda_i \quad p = 0, 1, \dots, M-1 \\ A(p) &= \left[\frac{1}{M-p} \sum_{i=p+1}^M \lambda_i \right]^{M-p} \\ E(p) &= \frac{1}{2} p(2M-p) \log N \end{aligned} \quad (12.5.46)$$

N : number of samples used to estimate the M autocorrelation lags

Some results on the quality of this order selection criterion are given in the paper by Wax and Kailath (1985). The MDL criterion is guaranteed to be consistent.

12.5.6 Experimental Results

In this section we illustrate with an example, the resolution characteristics of the eigenanalysis-based spectral estimation algorithms and compare their performance with the model-based methods and nonparametric methods. The signal sequence is

$$x(n) = \sum_{i=1}^4 A_i e^{j(2\pi f_i n + \phi_i)} + w(n)$$

where $A_i = 1$, $i = 1, 2, 3, 4$, $\{\phi_i\}$ are statistically independent random variables uniformly distributed on $(0, 2\pi)$, $\{w(n)\}$ is a zero-mean, white noise sequence with variance σ_w^2 , and the frequencies are $f_1 = -0.222$, $f_2 = -0.166$, $f_3 = 0.10$, and $f_4 = 0.122$. The sequence $\{x(n), 0 \leq n \leq 1023\}$ is used to estimate the number of frequency components and the corresponding values of their frequencies for $\sigma_w^2 = 0.1, 0.5, 1.0$, and $M = 12$ (length of the estimated autocorrelation).

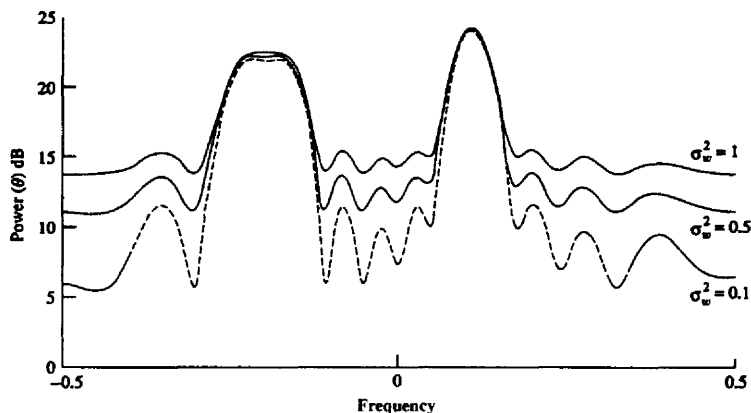


Figure 12.19 Power spectrum estimates from Blackman-Tukey method.

Figures 12.19, 12.20, 12.21, and 12.22 illustrate the estimated power spectra of the signal using the Blackman-Tukey method, the minimum variance method of Capon, the AR Yule-Walker method, and the MUSIC algorithm, respectively. The results from the ESPRIT algorithm are given in Table 12.2. From these results it is apparent that (1) the Blackman-Tukey method does not provide sufficient

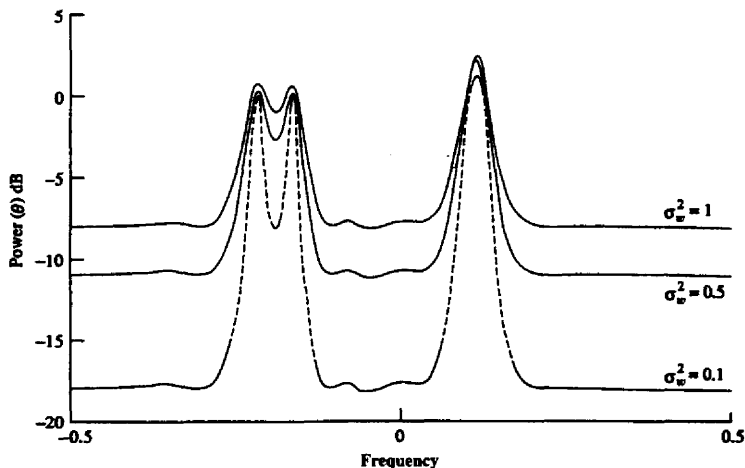


Figure 12.20 Power spectrum estimates from minimum variance method.

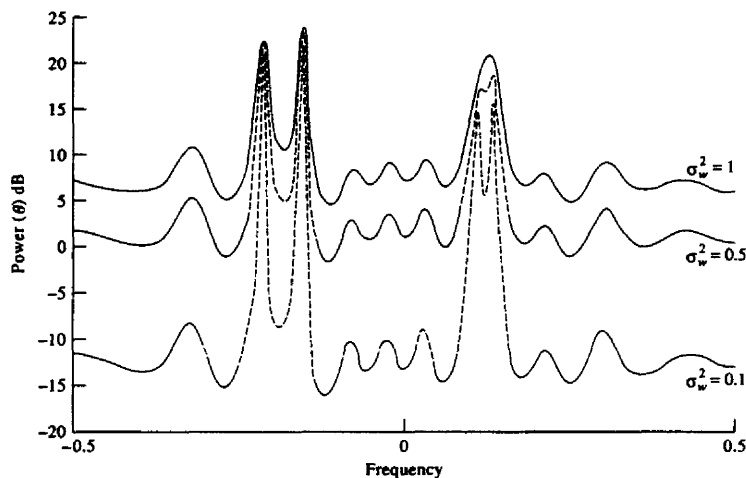


Figure 12.21 Power spectrum estimates from Yule-Walker AR method.

resolution to estimate the sinusoids from the data; (2) the minimum variance method of Capon resolves only the frequencies f_1 , f_2 but not f_3 and f_4 ; (3) the AR methods resolve all frequencies for $\sigma_w^2 = 0.1$ and $\sigma_w^2 = 0.5$; and (4) the MUSIC and ESPRIT algorithms not only recover all four sinusoids, but their performance for different values of σ_w^2 is essentially indistinguishable. We further observe that

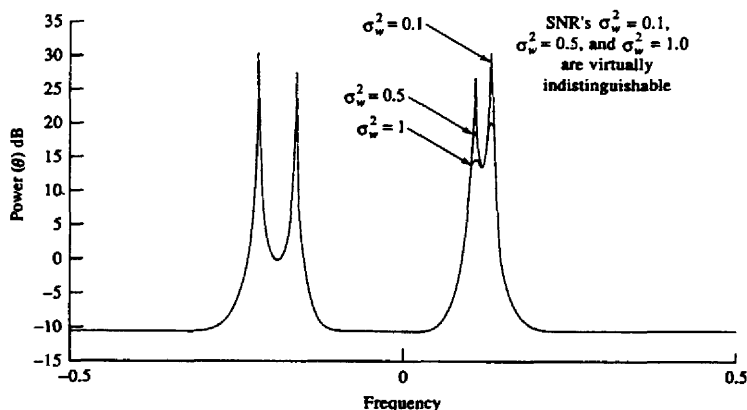


Figure 12.22 Power spectrum estimates from MUSIC algorithm.

TABLE 12.2 ESPRIT ALGORITHM

σ_w^2	$\hat{f}_1$	$\hat{f}_2$	$\hat{f}_3$	$\hat{f}_4$
0.1	-0.2227	-0.1668	-0.1224	-0.10071
0.5	-0.2219	-0.167	-0.121	0.0988
1.0	-0.222	-0.167	0.1199	0.1013
True values	-0.222	-0.166	0.122	0.100

the resolution properties of the minimum variance method and the AR method is a function of the noise variance. These results clearly demonstrate the power of the eigenanalysis-based algorithms in resolving sinusoids in additive noise.

In conclusion, we should emphasize that the high-resolution, eigenanalysis-based spectral estimation methods described in this section, namely MUSIC and ESPRIT, are not only applicable to sinusoidal signals, but apply more generally to the estimation of narrowband signals.

12.6 SUMMARY AND REFERENCES

Power spectrum estimation is one of the most important areas of research and applications in digital signal processing. In this chapter we have described the most important power spectrum estimation techniques and algorithms that have been developed over the past century, beginning with the nonparametric or classical methods based on the periodogram and concluding with the more modern parametric methods based on AR, MA, and ARMA linear models. Our treatment is limited in scope to single-time-series spectrum estimation methods, based on second moments (autocorrelation) of the statistical data.

The parametric and nonparametric methods that we described have been extended to multichannel and multidimensional spectrum estimation. The tutorial paper by McClellan (1982) treats the multidimensional spectrum estimation problem, while the paper by Johnson (1982) treats the multichannel spectrum estimation problem. Additional spectrum estimation methods have been developed for use with higher-order cumulants that involve the bispectrum and the trispectrum. A tutorial paper on these topics has been published by Nikias and Raghuveer (1987).

As evidenced from our previous discussion, power spectrum estimation is an area that has attracted many researchers and, as a result, thousands of papers have been published in the technical literature on this subject. Much of this work has been concerned with new algorithms and techniques, and modifications of existing techniques. Other work has been concerned with obtaining an understanding of the capabilities and limitations of the various power spectrum methods. In this context the statistical properties and limitations of the classical nonparametric methods have been thoroughly analyzed and are well understood. The parametric methods have also been investigated by many researchers, but the analysis of their

performance is difficult and, consequently, fewer results are available. Some of the papers that have addressed the problem of performance characteristics of parametric methods are those of Kromer (1969), Lacoss (1971), Berk (1974), Baggeroer (1976), Sakai (1979), Swingler (1980), Lang and McClellan (1980), and Tufts and Kumaresan (1982).

In addition to the references already given in this chapter on the various methods for spectrum estimation and their performance, we should include for reference some of the tutorial and survey papers. In particular, we cite the tutorial paper by Kay and Marple (1981), which includes about 280 references, the paper by Brillinger (1974), and the Special Issue on Spectral Estimation of the *IEEE Proceedings*, September 1982. Another indication of the widespread interest in the subject of spectrum estimation and analysis is the recent publication of texts by Gardner (1987), Kay (1988), and Marple (1987), and the IEEE books edited by Childers (1978) and Kesler (1986).

Many computer programs as well as software packages that implement the various spectrum estimation methods described in this chapter are available. One software package is available through the IEEE (*Programs for Digital Signal Processing*, IEEE Press, 1979); others are available commercially.

PROBLEMS

- 12.1 (a) By expanding (12.1.23), taking the expected value, and finally taking the limit as $T_0 \rightarrow \infty$, show that the right-hand side converges to $\Gamma_{xx}(F)$.
 (b) Prove that

$$\sum_{m=-N}^N r_{xx}(m) e^{-j2\pi f m} = \frac{1}{N} \left| \sum_{n=0}^{N-1} x(n) e^{-j2\pi f n} \right|^2$$

- 12.2 For zero mean, jointly Gaussian random variables, X_1, X_2, X_3, X_4 , it is well known [see Papoulis (1984)] that

$$E(X_1 X_2 X_3 X_4) = E(X_1 X_2) E(X_3 X_4) + E(X_1 X_3) E(X_2 X_4) + E(X_1 X_4) E(X_2 X_3)$$

Use this result to derive the mean-square value of $r'_{xx}(m)$, given by (12.1.27) and the variance, which is

$$\text{var}[r'_{xx}(m)] = E[|r'_{xx}(m)|^2] - |E[r'_{xx}(m)]|^2$$

- 12.3 By use of the expression for the fourth joint moment for Gaussian random variables, show that

$$\begin{aligned} \text{(a)} \quad E[P_{xx}(f_1) P_{xx}(f_2)] &= \sigma_x^4 \left\{ 1 + \left[\frac{\sin \pi(f_1 + f_2)N}{N \sin \pi(f_1 + f_2)} \right]^2 \right. \\ &\quad \left. + \left[\frac{\sin \pi(f_1 - f_2)N}{N \sin \pi(f_1 - f_2)} \right]^2 \right\} \end{aligned}$$

- (b) $\text{cov}[P_{xx}(f_1)P_{xx}(f_2)] = \sigma_x^4 \left\{ \left[\frac{\sin \pi(f_1 + f_2)N}{N \sin \pi(f_1 + f_2)} \right]^2 + \left[\frac{\sin \pi(f_1 - f_2)N}{N \sin \pi(f_1 - f_2)} \right]^2 \right\}$
- (c) $\text{var}[P_{xx}(f)] = \sigma_x^4 \left\{ 1 + \left(\frac{\sin 2\pi f N}{N \sin 2\pi f} \right)^2 \right\}$ under the condition that the sequence $x(n)$ is a zero-mean white Gaussian noise sequence with variance σ_x^2 .

12.4 Generalize the results in Problem 12.3 to a zero-mean Gaussian noise process with power density spectrum $\Gamma_{xx}(f)$. Then derive the variance of the periodogram $P_{xx}(f)$, as given by (12.1.38). (Hint: Assume that the colored Gaussian noise process is the output of a linear system excited by white Gaussian noise. Then use the appropriate relations given in Appendix A.)

12.5 Show that the periodogram values at frequencies $f_k = k/L$, $k = 0, 1, \dots, L-1$, given by (12.1.41) can be computed by passing the sequence through a bank of N IIR filters, where each filter has an impulse response

$$h_k(n) = e^{-j2\pi nk/N} u(n)$$

and then compute the magnitude-squared value of the filter outputs at $n = N$. Note that each filter has a pole on the unit circle at the frequency f_k .

12.6 Prove that the normalization factor given by (12.2.12) ensures that (12.2.19) is satisfied.

12.7 Let us consider the use of the DFT (computed via the FFT algorithm) to compute the autocorrelation of the complex-valued sequence $x(n)$, that is,

$$r_{xx}(m) = \frac{1}{N} \sum_{n=0}^{N-m-1} x^*(n)x(n+m), \quad m \geq 0$$

Suppose the size M of the FFT is much smaller than that of the data length N . Specifically, assume that $N = KM$.

- (a) Determine the steps needed to section $x(n)$ and compute $r_{xx}(m)$ for $-(M/2) + 1 \leq m \leq (M/2) - 1$, by using $4K$ M -point DFTs and one M -point IDFT.
- (b) Now consider the following three sequences $x_1(n)$, $x_2(n)$, and $x_3(n)$, each of duration M . Let the sequences $x_1(n)$ and $x_2(n)$ have arbitrary values in the range $0 \leq n \leq (M/2) - 1$, but are zero for $(M/2) \leq n \leq M - 1$. The sequence $x_3(n)$ is defined as

$$x_3(n) = \begin{cases} x_1(n), & 0 \leq \frac{M}{2} - 1 \\ x_2\left(n - \frac{M}{2}\right), & \frac{M}{2} \leq n \leq M - 1 \end{cases}$$

Determine a simple relationship among the M -point DFTs $X_1(k)$, $X_2(k)$, and $X_3(k)$.

- (c) By using the result in part (b), show how the computation of the DFTs in part (a) can be reduced in number from $4K$ to $2K$.

12.8 The Bartlett method is used to estimate the power spectrum of a signal $x(n)$. We know that the power spectrum consists of a single peak with a 3-dB bandwidth of 0.01 cycle per sample, but we do not know the location of the peak.

- (a) Assuming that N is large, determine the value of $M = N/K$ so that the spectral window is narrower than the peak.
 (b) Explain why it is not advantageous to increase M beyond the value obtained in part (a).

12.9 Suppose we have $N = 1000$ samples from a sample sequence of a random process.

- (a) Determine the frequency resolution of the Bartlett, Welch (50% overlap), and Blackman–Tukey methods for a quality factor $Q = 10$.
 (b) Determine the record lengths (M) for the Bartlett, Welch (50% overlap), and Blackman–Tukey methods.

12.10 Consider the problem of continuously estimating the power spectrum from a sequence $x(n)$ based on averaging periodograms with exponential weighting into the past. Thus with $P_{xx}^{(0)}(f) = 0$, we have

$$P_{xx}^{(m)}(f) = w P_{xx}^{(m-1)}(f) + \frac{1-w}{M} \left| \sum_{n=0}^{M-1} x_m(n) e^{-j2\pi f n} \right|^2$$

where successive periodograms are assumed to be uncorrelated and w is the (exponential) weighting factor.

- (a) Determine the mean and variance of $P_{xx}^{(m)}(f)$ for a Gaussian random process.
 (b) Repeat the analysis of part (a) for the case in which the modified periodogram defined by Welch is used in the averaging with no overlap.

12.11 The periodogram in the Bartlett method can be expressed as

$$P_{xx}^{(i)}(f) = \sum_{m=-(M-1)}^{M-1} \left(1 - \frac{|m|}{M} \right) r_{xx}^{(i)}(m) e^{-j2\pi f m}$$

where $r_{xx}^{(i)}(m)$ is the estimated autocorrelation sequence obtained from the i th block of data. Show that $P_{xx}^{(i)}(f)$ can be expressed as

$$P_{xx}^{(i)}(f) = E^w(f) R_{xx}^{(i)}(f)$$

where

$$E(f) = [1 \quad e^{j2\pi f} \quad e^{j4\pi f} \quad \dots \quad e^{j2\pi(M-1)f}]^T$$

and therefore,

$$P_{xx}^B(f) = \frac{1}{K} \sum_{k=1}^K E^w(f) R_{xx}^{(k)}(f) E(f)$$

12.12 Derive the recursive order-update equation given in (12.3.19).

12.13 Determine the mean and the autocorrelation of the sequence $x(n)$, which is the output of a ARMA (1, 1) process described by the difference equation

$$x(n) = \frac{1}{2}x(n-1) + w(n) - w(n-1)$$

where $w(n)$ is a white noise process with variance σ_w^2 .

12.14 Determine the mean and the autocorrelation of the sequence $x(n)$ generated by the MA(2) process described by the difference equation

$$x(n) = w(n) - 2w(n-1) + w(n-2)$$

where $w(n)$ is a white noise process with variance σ_w^2 .

12.15 An MA(2) process has the autocorrelation sequence

$$\gamma_{xx}(m) = \begin{cases} 6\sigma_w^2, & m = 0 \\ -4\sigma_w^2, & m = \pm 1 \\ -2\sigma_w^2, & m = \pm 2 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Determine the coefficients of the MA(2) process that have the foregoing autocorrelation.
 (b) Is the solution unique? If not, give all the possible solutions.

12.16 An MA(2) process has the autocorrelation sequence

$$\gamma_{xx}(m) = \begin{cases} \sigma_w^2, & m = 0 \\ -\frac{35}{62}\sigma_w^2, & m = \pm 1 \\ \frac{6}{62}\sigma_w^2, & m = \pm 2 \end{cases}$$

- (a) Determine the coefficients of the minimum-phase system for the MA(2) process.
 (b) Determine the coefficients of the maximum-phase system for the MA(2) process.
 (c) Determine the coefficients of the mixed-phase system for the MA(2) process.

12.17 Consider the linear system described by the difference equation

$$y(n) = 0.8y(n-1) + x(n) + x(n-1)$$

where $x(n)$ is a wide-sense stationary random process with zero mean and autocorrelation

$$\gamma_{xx}(m) = \left(\frac{1}{2}\right)^{|m|}$$

- (a) Determine the power density spectrum of the output $y(n)$.
 (b) Determine the autocorrelation $\gamma_{yy}(m)$ of the output.
 (c) Determine the variance σ_y^2 of the output.

12.18 From (12.3.6) and (12.3.9) we note that an AR(p) stationary random process satisfies the equation

$$\gamma_{xx}(m) + \sum_{k=1}^p a_p(k)\gamma_{xx}(m-k) = \begin{cases} \sigma_w^2, & m = 0, \\ 0, & 1 \leq m \leq p, \end{cases}$$

where $a_p(k)$ are the prediction coefficients of the linear predictor of order p and σ_w^2 is the minimum mean-square prediction error. If the $(p+1) \times (p+1)$ autocorrelation matrix Γ_{xx} in (12.3.9) is positive definite, prove that:

- (a) The reflection coefficients $|K_m| < 1$ for $1 \leq m \leq p$.
 (b) The polynomial

$$A_p(z) = 1 + \sum_{k=1}^p a_p(k)z^{-k}$$

has all its roots inside the unit circle (i.e., it is minimum phase).

12.19 Consider the AR(3) process generated by the equation

$$x(n) = \frac{14}{24}x(n-1) + \frac{9}{24}x(n-2) - \frac{1}{24}x(n-3) + w(n)$$

where $w(n)$ is a stationary white noise process with variance σ_w^2 .

- (a) Determine the coefficients of the optimum $p = 3$ linear predictor.
- (b) Determine the autocorrelation sequence $\gamma_{xx}(m)$, $0 \leq m \leq 5$.
- (c) Determine the reflection coefficients corresponding to the $p = 3$ linear predictor.

12.20* An AR(2) process is described by the difference equation

$$x(n) = 0.81x(n-2) + w(n)$$

where $w(n)$ is a white noise process with variance σ_w^2 .

- (a) Determine the parameters of the MA(2), MA(4), and MA(8) models which provide a minimum mean-square error fit to the data $x(n)$.
- (b) Plot the true spectrum and those of the MA(q), $q = 2, 4, 8$, spectra and compare the results. Comment on how well the MA(q) models approximate the AR(2) process.

12.21 An MA(2) process is described by the difference equation

$$x(n) = w(n) + 0.81w(n-2)$$

where $w(n)$ is a white noise process with variance σ_w^2 .

- (a) Determine the parameters of the AR(2), AR(4), and AR(8) models that provide a minimum mean-square error fit to the data $x(n)$.
- (b) Plot the true spectra m and those of the AR(p), $p = 2, 4, 8$, and compare the results. Comment on how well the AR(p) models approximate the MA(2) process.

12.22 The z -transform of the autocorrelation $\gamma_{xx}(m)$ of an ARMA(1, 1) process is

$$\Gamma_{xx}(z) = \sigma_w^2 H(z)H(z^{-1})$$

$$\Gamma_{xx}(z) = \frac{4\sigma_w^2}{9} \frac{5 - 2z - 2z^{-1}}{10 - 3z^{-1} - 3z}$$

- (a) Determine the minimum-phase system function $H(z)$.
- (b) Determine the system function $H(z)$ for a mixed-phase stable system.

12.23 Consider a FIR filter with coefficient vector

$$[1 \quad -2r \cos \theta \quad r^2]$$

- (a) Determine the reflection coefficients for the corresponding FIR lattice filter.
- (b) Determine the values of the reflection coefficients in the limit as $r \rightarrow 1$.

12.24 An AR(3) process is characterized by the prediction coefficients

$$a_3(1) = -1.25, \quad a_3(2) = 1.25, \quad a_3(3) = -1$$

- (a) Determine the reflection coefficients.
- (b) Determine $\gamma_{xx}(m)$ for $0 \leq m \leq 3$.
- (c) Determine the mean-square prediction error.

12.25 The autocorrelation sequence for a random process is

$$\gamma_{xx}(m) = \begin{cases} 1, & m = 0 \\ -0.5, & m = \pm 1 \\ 0.625, & m = \pm 2 \\ -0.6875, & m = \pm 3 \\ 0 & \text{otherwise} \end{cases}$$

Determine the system functions $A_m(z)$ for the prediction-error filters for $m = 1, 2, 3$, the reflection coefficients $\{K_m\}$, and the corresponding mean-square prediction errors.

- 12.26 (a)** Determine the power spectra for the random processes generated by the following difference equations.

$$(1) \quad x(n) = -0.81x(n-2) + w(n) - w(n-1)$$

$$(2) \quad x(n) = w(n) - w(n-2)$$

$$(3) \quad x(n) = -0.81x(n-2) + w(n)$$

where $w(n)$ is a white noise process with variance σ_w^2 .

- (b) Sketch the spectra for the processes given in part (a).

- (c) Determine the autocorrelation $\gamma_{xx}(m)$ for the processes in (2) and (3).

- 12.27** The autocorrelation sequence for an AR process $x(n)$ is

$$\gamma_{xx}(m) = \left(\frac{1}{4}\right)^{|m|}$$

- (a) Determine the difference equation for $x(n)$.

- (b) Is your answer unique? If not, give any other possible solutions.

- 12.28** Repeat Problem 12.27 for an AR process with autocorrelation

$$\gamma_{xx}(m) = a^{|m|} \cos \frac{\pi m}{2}$$

where $0 < a < 1$.

- 12.29** The Bartlett method is used to estimate the power spectrum of a signal from a sequence $x(n)$ consisting of $N = 2400$ samples.

- (a) Determine the smallest length M of each segment in the Bartlett method that yields a frequency resolution of $\Delta f = 0.01$.

- (b) Repeat part (a) for $\Delta f = 0.02$.

- (c) Determine the quality factors Q_B for parts (a) and (b).

- 12.30** Prove that a FIR filter with system function

$$A_p(z) = 1 + \sum_{k=1}^p a_p(k)z^{-k}$$

and reflection coefficients $|K_k| < 1$ for $1 \leq k \leq p-1$ and $|K_p| > 1$ is maximum phase [all the roots of $A_p(z)$ lie outside the unit circle].

- 12.31** A random process $x(n)$ is characterized by the power density spectrum

$$\Gamma_{xx}(f) = \sigma_w^2 \frac{|e^{j2\pi f} - 0.9|^2}{|e^{j2\pi f} - j0.9|^2 |e^{j2\pi f} + j0.9|^2}$$

where σ_w^2 is a constant (scale factor).

- (a) If we view $\Gamma_{xx}(f)$ as the power spectrum at the output of a linear pole-zero system $H(z)$ driven by white noise, determine $H(z)$.

- (b) Determine the system function of a stable system (noise-whitening filter) that produces a white noise output when excited by $x(n)$.

- 12.32** The N -point DFT of a random sequence $x(n)$ is

$$X(k) = \sum_{n=0}^{N-1} x(n)e^{-j2\pi nk/N}$$

Assume that $E[x(n)] = 0$ and $E[x(n)x(n+m)] = \sigma_x^2 \delta(m)$ [i.e., $x(n)$ is a white noise process].

- (a) Determine the variance of $X(k)$.
 (b) Determine the autocorrelation of $X(k)$.

12.33 Suppose that we represent an ARMA(p, q) process as a cascade of a MA(q) followed by an AR(p) model. The input-output equation for the MA(q) model is

$$v(n) = \sum_{k=0}^q b_k w(n-k)$$

where $w(n)$ is a white noise process. The input-output equation for the AR(p) model is

$$x(n) + \sum_{k=1}^p a_k x(n-k) = v(n)$$

- (a) By computing the autocorrelation of $v(n)$, show that

$$\gamma_{vv}(m) = \sigma_w^2 \sum_{k=0}^{q-m} b_k^* b_{k+m}$$

- (b) Show that

$$\gamma_{vv}(m) = \sum_{k=0}^p a_k \gamma_{vx}(m+k) \quad a_0 = 1$$

where $\gamma_{vx}(m) = E[v(n+m)x^*(n)]$.

12.34 Determine the autocorrelation $\gamma_{xx}(m)$ of the random sequence

$$x(n) = A \cos(\omega_1 n + \phi)$$

where the amplitude A and the frequency ω_1 are (known) constants and ϕ is a uniformly distributed random phase over the interval $(0, 2\pi)$.

12.35 Suppose that the AR(2) process in Problem 12.20 is corrupted by an additive white noise process $v(n)$ with variance σ_v^2 . Thus we have

$$y(n) = x(n) + v(n)$$

- (a) Determine the difference equation for $y(n)$ and thus demonstrate that $y(n)$ is an ARMA(2, 2) process. Determine the coefficients of the ARMA process.
 (b) Generalize the result in part (a) to an AR(p) process

$$x(n) = - \sum_{k=1}^p a_k x(n-k) + w(n)$$

and

$$y(n) = x(n) + v(n)$$

12.36 (a) Determine the autocorrelation of the random sequence

$$x(n) = \sum_{k=1}^K A_k \cos(\omega_k n + \phi_k) + w(n)$$

where $\{A_k\}$ are constant amplitudes, $\{\omega_k\}$ are constant frequencies, and $\{\phi_k\}$ are mutually statistically independent and uniformly distributed random phases. The noise sequence $w(n)$ is white with variance σ_w^2 .

- (b) Determine the power density spectrum of $x(n)$.

- 12.37** The harmonic decomposition problem considered by Pisarenko can be expressed as the solution to the equation

$$\mathbf{a}^H \Gamma_{yy} \mathbf{a} = \sigma_w^2 \mathbf{a}^H \mathbf{a}$$

The solution for $\mathbf{a}$ can be obtained by minimizing the quadratic form $\mathbf{a}^H \Gamma_{yy} \mathbf{a}$ subject to the constraint that $\mathbf{a}^H \mathbf{a} = 1$. The constraint can be incorporated into the performance index by means of a Lagrange multiplier. Thus the performance index becomes

$$\mathcal{E} = \mathbf{a}^H \Gamma_{yy} \mathbf{a} + \lambda(1 - \mathbf{a}^H \mathbf{a})$$

By minimizing $\mathcal{E}$ with respect to $\mathbf{a}$, show that this formulation is equivalent to the Pisarenko eigenvalue problem given in (12.5.9) with the Lagrange multiplier playing the role of the eigenvalue. Thus show that the minimum of $\mathcal{E}$ is the minimum eigenvalue σ_w^2 .

- 12.38** The autocorrelation of a sequence consisting of a sinusoid with random phase in noise is

$$\gamma_{xx}(m) = P \cos 2\pi f_1 m + \sigma_w^2 \delta(m)$$

where f_1 is the frequency of the sinusoid, P is its power, and σ_w^2 is the variance of the noise. Suppose that we attempt to fit an AR(2) model to the data.

- Determine the optimum coefficients of the AR(2) model as a function of σ_w^2 and f_1 .
 - Determine the reflection coefficients K_1 and K_2 corresponding to the AR(2) model parameters.
 - Determine the limiting values of the AR(2) parameters and (K_1, K_2) as $\sigma_w^2 \rightarrow 0$.
- 12.39** This problem involves the use of crosscorrelation to detect a signal in noise and estimate the time delay in the signal. A signal $x(n)$ consists of a pulsed sinusoid corrupted by a stationary zero-mean white noise sequence. That is,

$$x(n) = y(n - n_0) + w(n) \quad 0 \leq n \leq N - 1$$

where $w(n)$ is the noise with variance σ_w^2 and the signal is

$$y(n) = \begin{cases} A \cos \omega_0 n, & 0 \leq n \leq M - 1 \\ 0, & \text{otherwise} \end{cases}$$

The frequency ω_0 is known but the delay n_0 , which is a positive integer, is unknown, and is to be determined by crosscorrelating $x(n)$ with $y(n)$. Assume that $N > M + n_0$. Let

$$r_{xy}(m) = \sum_{n=0}^{N-1} y(n-m)x(n)$$

denote the crosscorrelation sequence between $x(n)$ and $y(n)$. In the absence of noise this function exhibits a peak at delay $m = n_0$. Thus n_0 is determined with no error. The presence of noise can lead to errors in determining the unknown delay.

- For $m = n_0$, determine $E[r_{xy}(n_0)]$. Also, determine the variance, $\text{var}[r_{xy}(n_0)]$, due to the presence of the noise. In both calculations, assume that the double frequency term averages to zero. That is, $M \gg 2\pi/\omega_0$.
- Determine the signal-to-noise ratio, defined as

$$\text{SNR} = \frac{\{E[r_{xy}(n_0)]\}^2}{\text{var}[r_{xy}(n_0)]}$$

(c) What is the effect of the pulse duration M on the SNR?

12.40* Generate 100 samples of a zero-mean white noise sequence $w(n)$ with variance $\sigma_w^2 = \frac{1}{12}$, by using a uniform random number generator.

(a) Compute the autocorrelation of $w(n)$ for $0 \leq m \leq 15$.

(b) Compute the periodogram estimate $P_{xx}(f)$ and plot it.

(c) Generate 10 different realizations of $w(n)$ and compute the corresponding sample autocorrelation sequences $r_k(m)$, $1 \leq k \leq 10$ and $0 \leq m \leq 15$.

(d) Compute and plot the average periodogram for part (c):

$$r_{av}(m) = \frac{1}{10} \sum_{k=1}^{10} r_k(m)$$

(e) Comment on the results in parts (a) through (d).

12.41* A random signal is generated by passing zero-mean white Gaussian noise with unit variance through a filter with system function

$$H(z) = \frac{1}{(1 + az^{-1} + 0.99z^{-2})(1 - az^{-1} + 0.98z^{-2})}$$

(a) Sketch a typical plot of the theoretical power spectrum $\Gamma_{xx}(f)$ for a small value of the parameter a (i.e., $0 < a < 0.1$). Pay careful attention to the value of the two spectral peaks and the value of $P_{xx}(\omega)$ for $\omega = \pi/2$.

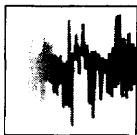
(b) Let $a = 0.1$. Determine the section length M required to resolve the spectral peaks of $\Gamma_{xx}(f)$ when using Bartlett's method.

(c) Consider the Blackman-Tukey method of smoothing the periodogram. How many lags of the correlation estimate must be used to obtain resolution comparable to that of the Bartlett estimate considered in part (b)? How many data must be used if the variance of the estimate is to be comparable to that of a four-section Bartlett estimate?

(d) For $a = 0.05$, fit an AR(4) model to 100 samples of the data based on the Yule-Walker method and plot the power spectrum. Avoid transient effects by discarding the first 200 samples of the data.

(e) Repeat part (d) with the Burg method.

(f) Repeat parts (d) and (e) for 50 data samples and comment on similarities and differences in the results.



Appendix A

Random Signals, Correlation Functions, and Power Spectra

In this appendix we provide a brief review of the characterization of random signals in terms of statistical averages expressed in both the time domain and the frequency domain. The reader is assumed to have a background in probability theory and random processes, at the level given in the books of Helstrom (1990) and Peebles (1987).

Random Processes

Many physical phenomena encountered in nature are best characterized in statistical terms. For example, meteorological phenomena such as air temperature and air pressure fluctuate randomly as a function of time. Thermal noise voltages generated in the resistors of electronic devices, such as a radio or television receiver, are also randomly fluctuating phenomena. These are just a few examples of random signals. Such signals are usually modeled as infinite-duration infinite-energy signals.

Suppose that we take the set of waveforms corresponding to the air temperature in different cities around the world. For each city there is a corresponding waveform that is a function of time, as illustrated in Fig. A.1. The set of all possible waveforms is called an *ensemble* of time functions or, equivalently, a *random process*. The waveform for the temperature in any particular city is a *single realization* or a *sample function* of the random process.

Similarly, the thermal noise voltage generated in a resistor is a single realization or a sample function of the random process consisting of all noise voltage waveforms generated by the set of all resistors.

The set (ensemble) of all possible noise waveforms of a random process is denoted as $X(t, S)$, where t represents the time index and S represents the set (sample space) of all possible sample functions. A single waveform in the ensemble is denoted by $x(t, s)$. Usually, we drop the variable s (or S) for notational convenience, so that the random process is denoted as $X(t)$ and a single realization is denoted as $x(t)$.

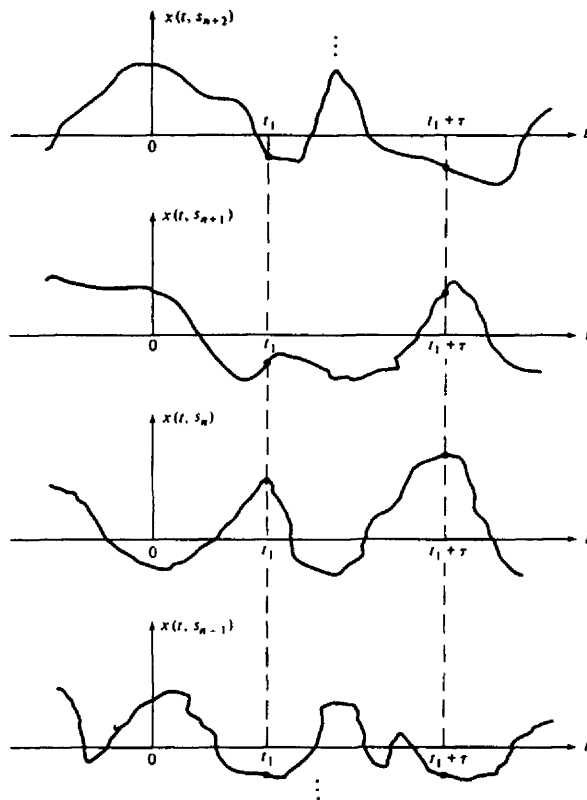


Figure A.1

Having defined a random process $X(t)$ as an ensemble of sample functions, let us consider the values of the process for any set of time instants $t_1 > t_2 > \cdots > t_n$, where n is any positive integer. In general, the samples $X_{t_i} \equiv x(t_i)$, $i = 1, 2, \dots, n$ are n random variables characterized statistically by their joint probability density function (PDF) denoted as $p(x_{t_1}, x_{t_2}, \dots, x_{t_n})$ for any n .

Stationary Random Processes

Suppose that we have n samples of the random process $X(t)$ at $t = t_i$, $i = 1, 2, \dots, n$, and another set of n samples displaced in time from the first set by an amount τ . Thus the second set of samples are $X_{t_i+\tau} \equiv X(t_i + \tau)$, $i = 1, 2, \dots, n$, as

shown in Fig. A.1. This second set of n random variables is characterized by the joint probability density function $p(x_{i_1+\tau}, \dots, x_{i_n+\tau})$. The joint PDFs of the two sets of random variables may or may not be identical. When they are identical, then

$$p(x_{i_1}, x_{i_2}, \dots, x_{i_n}) = p(x_{i_1+\tau}, x_{i_2+\tau}, \dots, x_{i_n+\tau}) \quad (\text{A.1})$$

for all τ and all n , then the random process is said to be *stationary in the strict sense*. In other words, the statistical properties of a stationary random process are invariant to a translation of the time axis. On the other hand, when the joint PDFs are different, the random process is nonstationary.

Statistical (Ensemble) Averages

Let us consider a random process $X(t)$ sampled at time instant $t = t_i$. Thus $X(t_i)$ is a random variable with PDF $p(x_{t_i})$. The l th *moment* of the random variable is defined as the *expected value* of $X^l(t_i)$, that is,

$$E(x_{t_i}^l) = \int_{-\infty}^{\infty} x_{t_i}^l p(x_{t_i}) dx_{t_i} \quad (\text{A.2})$$

In general, the value of the l th moment depends on the time instant t_i , if the PDF of X_{t_i} depends on t_i . When the process is stationary, however, $p(x_{t_i+\tau}) = p(x_{t_i})$ for all τ . Hence the PDF is independent of time and, consequently, the l th moment is independent of time (a constant).

Next, let us consider the two random variables $X_{t_i} = X(t_i)$, $i = 1, 2$, corresponding to samples of $X(t)$ taken at $t = t_1$ and $t = t_2$. The statistical (ensemble) correlation between X_{t_1} and X_{t_2} is measured by the joint moment

$$E(X_{t_1} X_{t_2}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_{t_1} x_{t_2} p(x_{t_1}, x_{t_2}) dx_{t_1} dx_{t_2} \quad (\text{A.3})$$

Since the joint moment depends on the time instants t_1 and t_2 , it is denoted as $\gamma_{xx}(t_1, t_2)$ and is called the *autocorrelation function* of the random process. When the process $X(t)$ is stationary, the joint PDF of the pair (X_{t_1}, X_{t_2}) is identical to the joint PDF of the pair $(X_{t_1+\tau}, X_{t_2+\tau})$ for any arbitrary τ . This implies that the autocorrelation function of $X(t)$ depends on the time difference $t_1 - t_2 = \tau$. Hence for a stationary real-valued random process the autocorrelation function is

$$\gamma_{xx}(\tau) = E[X_{t_1+\tau} X_{t_1}] \quad (\text{A.4})$$

On the other hand,

$$\gamma_{xx}(-\tau) = E(X_{t_1-\tau} X_{t_1}) = E(X_{t_1} X_{t_1+\tau}) = \gamma_{xx}(\tau) \quad (\text{A.5})$$

Therefore, $\gamma_{xx}(\tau)$ is an even function. We also note that $\gamma_{xx}(0) = E(X_{t_1}^2)$ is the *average power* of the random process.

There exist nonstationary processes with the property that the mean value of the process is a constant and the autocorrelation function satisfies the property $\gamma_{xx}(t_1, t_2) = \gamma_{xx}(t_1 - t_2)$. Such a process is called *wide-sense stationary*. Clearly,

wide-sense stationarity is a less stringent condition than strict-sense stationarity. In our treatment we shall require only that the processes be wide-sense stationary.

Related to the autocorrelation function is the autocovariance function, which is defined as

$$\begin{aligned} c_{xx}(t_1, t_2) &= E\{[X_{t_1} - m(t_1)][X_{t_2} - m(t_2)]\} \\ &= \gamma_{xx}(t_1, t_2) - m(t_1)m(t_2) \end{aligned} \quad (\text{A.6})$$

where $m(t_1) = E(X_{t_1})$ and $m(t_2) = E(X_{t_2})$ are the mean values of X_{t_1} and X_{t_2} , respectively. When the process is stationary,

$$c_{xx}(t_1, t_2) = c_{xx}(t_1 - t_2) = c_{xx}(\tau) = \gamma_{xx}(\tau) - m_x^2 \quad (\text{A.7})$$

where $\tau = t_1 - t_2$. Furthermore, the variance of the process is $\sigma_x^2 = c_{xx}(0) = \gamma_{xx}(0) - m_x^2$.

Statistical Averages for Joint Random Processes

Let $X(t)$ and $Y(t)$ be two random processes and let $X_i \equiv X(t_i)$, $i = 1, 2, \dots, n$, and $Y_j \equiv Y(t'_j)$, $j = 1, 2, \dots, m$, represent the random variables at times $t_1 > t_2 > \dots > t_n$ and $t'_1 > t'_2 > \dots > t'_m$, respectively. The two sets of random variables are characterized statistically by the joint PDF

$$p(x_{t_1}, x_{t_2}, \dots, x_{t_n}, y_{t'_1}, y_{t'_2}, \dots, y_{t'_m})$$

for any set of time instants $\{t_i\}$ and $\{t'_j\}$ and for any positive integer values of m and n .

The *crosscorrelation function* of $X(t)$ and $Y(t)$, denoted as $\gamma_{xy}(t_1, t_2)$, is defined by the joint moment

$$\gamma_{xy}(t_1, t_2) \equiv E(X_{t_1} Y_{t_2}) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_{t_1} y_{t_2} p(x_{t_1}, y_{t_2}) dx_{t_1} dy_{t_2} \quad (\text{A.8})$$

and the crosscovariance is

$$c_{xy}(t_1, t_2) = \gamma_{xy}(t_1, t_2) - m_x(t_1)m_y(t_2) \quad (\text{A.9})$$

When the random processes are jointly and individually stationary, we have $\gamma_{xy}(t_1, t_2) = \gamma_{xy}(t_1 - t_2)$ and $c_{xy}(t_1, t_2) = c_{xy}(t_1 - t_2)$. In this case

$$\gamma_{xy}(-\tau) = E(X_{t_1} Y_{t_1+\tau}) = E(X_{t'_1-\tau} Y_{t'_1}) = \gamma_{yx}(\tau) \quad (\text{A.10})$$

The random processes $X(t)$ and $Y(t)$ are said to be statistically independent if and only if

$$p(x_{t_1}, x_{t_2}, \dots, x_{t_n}, y_{t'_1}, y_{t'_2}, \dots, y_{t'_m}) = p(x_{t_1}, \dots, x_{t_n})p(y_{t'_1}, \dots, y_{t'_m})$$

for all choices of t_i, t'_j and for all positive integers n and m . The processes are said to be uncorrelated if

$$\gamma_{xy}(t_1, t_2) = E(X_{t_1})E(Y_{t_2}) \quad (\text{A.11})$$

so that $c_{xy}(t_1, t_2) = 0$.

A complex-valued random process $Z(t)$ is defined as

$$Z(t) = X(t) + jY(t) \quad (\text{A.12})$$

where $X(t)$ and $Y(t)$ are random processes. The joint PDF of the complex-valued random variables $Z_i \equiv Z(t_i)$, $i = 1, 2, \dots$, is given by the joint PDF of the components (X_i, Y_i) , $i = 1, 2, \dots, n$. Thus the PDF that characterizes Z_i , $i = 1, 2, \dots, n$ is

$$p(x_{t_1}, x_{t_2}, \dots, x_{t_n}, y_{t_1}, y_{t_2}, \dots, y_{t_n})$$

A complex-valued random process $Z(t)$ is encountered in the representation of the in-phase and quadrature components of the lowpass equivalent of a narrowband random signal or noise. An important characteristic of such a process is its autocorrelation function, which is defined as

$$\begin{aligned} \gamma_{zz}(t_1, t_2) &= E(Z_{t_1} Z_{t_2}^*) \\ &= E[(X_{t_1} + jY_{t_1})(X_{t_2} - jY_{t_2})] \\ &= \gamma_{xx}(t_1, t_2) + \gamma_{yy}(t_1, t_2) + j[\gamma_{yx}(t_1, t_2) - \gamma_{xy}(t_1, t_2)] \end{aligned} \quad (\text{A.13})$$

When the random processes $X(t)$ and $Y(t)$ are jointly and individually stationary, the autocorrelation function of $Z(t)$ becomes

$$\gamma_{zz}(t_1, t_2) = \gamma_{zz}(t_1 - t_2) = \gamma_{zz}(\tau)$$

where $\tau = t_1 - t_2$. The complex conjugate of (A.13) is

$$\gamma_{zz}^*(\tau) = E(Z_{t_1}^* Z_{t_1-\tau}) = \gamma_{zz}(-\tau) \quad (\text{A.14})$$

Now, suppose that $Z(t) = X(t) + jY(t)$ and $W(t) = U(t) + jV(t)$ are two complex-valued random processes. Their crosscorrelation function is defined as

$$\begin{aligned} \gamma_{zw}(t_1, t_2) &= E(Z_{t_1} W_{t_2}^*) \\ &= E[(X_{t_1} + jY_{t_1})(U_{t_2} - jV_{t_2})] \\ &= \gamma_{xu}(t_1, t_2) + \gamma_{yv}(t_1, t_2) + j[\gamma_{yu}(t_1, t_2) - \gamma_{xv}(t_1, t_2)] \end{aligned} \quad (\text{A.15})$$

When $X(t)$, $Y(t)$, $U(t)$, and $V(t)$ are pairwise stationary, the crosscorrelation functions in (A.15) become functions of the time difference $\tau = t_1 - t_2$. In addition, we have

$$\gamma_{zw}^*(\tau) = E(Z_{t_1}^* W_{t_1-\tau}) = E(Z_{t_1+\tau}^* W_{t_1}) = \gamma_{wz}(-\tau) \quad (\text{A.16})$$

Power Density Spectrum

A stationary random process is an infinite-energy signal and hence its Fourier transform does not exist. The spectral characteristic of a random process is obtained according to the Wiener-Khinchine theorem, by computing the Fourier transform of the autocorrelation function. That is, the distribution of power with

frequency is given by the function

$$\Gamma_{xx}(F) = \int_{-\infty}^{\infty} \gamma_{xx}(\tau) e^{-j2\pi F\tau} d\tau \quad (\text{A.17})$$

The inverse Fourier transform is given as

$$\gamma_{xx}(\tau) = \int_{-\infty}^{\infty} \Gamma_{xx}(F) e^{j2\pi F\tau} dF \quad (\text{A.18})$$

We observe that

$$\begin{aligned} \gamma_{xx}(0) &= \int_{-\infty}^{\infty} \Gamma_{xx}(F) dF \\ &= E(X_t^2) \geq 0 \end{aligned} \quad (\text{A.19})$$

Since $E(X_t^2) = \gamma_{xx}(0)$ represents the average power of the random process, which is the area under $\Gamma_{xx}(F)$, it follows that $\Gamma_{xx}(F)$ is the distribution of power as a function of frequency. For this reason, $\Gamma_{xx}(F)$ is called the *power density spectrum* of the random process.

If the random process is real, $\gamma_{xx}(\tau)$ is real and even and hence $\Gamma_{xx}(F)$ is real and even. If the random process is complex valued, $\gamma_{xx}(\tau) = \gamma_{xx}^*(-\tau)$ and, hence

$$\begin{aligned} \Gamma_{xx}^*(F) &= \int_{-\infty}^{\infty} \gamma_{xx}^*(\tau) e^{j2\pi F\tau} d\tau = \int_{-\infty}^{\infty} \gamma_{xx}^*(-\tau) e^{-j2\pi F\tau} d\tau \\ &= \int_{-\infty}^{\infty} \gamma_{xx}(\tau) e^{-j2\pi F\tau} d\tau = \Gamma_{xx}(F) \end{aligned}$$

Therefore, $\Gamma_{xx}(F)$ is always real.

The definition of the power density spectrum can be extended to two jointly stationary random processes $X(t)$ and $Y(t)$, which have a crosscorrelation function $\gamma_{xy}(\tau)$. The Fourier transform of $\gamma_{xy}(\tau)$ is

$$\Gamma_{xy}(F) = \int_{-\infty}^{\infty} \gamma_{xy}(\tau) e^{-j2\pi F\tau} d\tau \quad (\text{A.20})$$

which is called the *cross-power density spectrum*. It is easily shown that $\Gamma_{xy}^*(F) = \Gamma_{yx}(-F)$. For real random processes, the condition is $\Gamma_{yx}(F) = \Gamma_{xy}(-F)$.

Discrete-Time Random Signals

This characterization of continuous-time random signals can be easily carried over to discrete-time signals. Such signals are usually obtained by uniformly sampling a continuous-time random process.

A discrete-time random process $X(n)$ consists of an ensemble of sample sequences $x(n)$. The statistical properties of $X(n)$ are similar to the characterization of $X(t)$, with the restriction that n is now an integer (time) variable. To be specific, we state the form for the important moments that we use in this text.

The l th moment of $X(n)$ is defined as

$$E(X_n^l) = \int_{-\infty}^{\infty} x_n^l p(x_n) dx_n \quad (\text{A.21})$$

and the autocorrelation sequence is

$$\gamma_{xx}(n, k) = E(X_n X_k) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_n x_k p(x_n, x_k) dx_n dx_k \quad (\text{A.22})$$

Similarly, the autocovariance is

$$c_{xx}(n, k) = \gamma_{xx}(n, k) - E(X_n)E(X_k) \quad (\text{A.23})$$

For a stationary process, we have the special forms ($m = n - k$)

$$\begin{aligned} \gamma_{xx}(n - k) &= \gamma_{xx}(m) \\ c_{xx}(n - k) &= c_{xx}(m) = \gamma_{xx}(m) - m_x^2 \end{aligned} \quad (\text{A.24})$$

where $m_x = E(X_n)$ is the mean of the random process. The variance is defined as $\sigma^2 = c_{xx}(0) = \gamma_{xx}(0) - m_x^2$.

For a complex-valued stationary process $Z(n) = X(n) + jY(n)$, we have

$$\gamma_{zz}(m) = \gamma_{xx}(m) + \gamma_{yy}(m) + j[\gamma_{yx}(m) - \gamma_{xy}(m)] \quad (\text{A.25})$$

and the crosscorrelation sequence of two complex-valued stationary sequences is

$$\gamma_{zw}(m) = \gamma_{xu}(m) + \gamma_{yv}(m) + j[\gamma_{yu}(m) - \gamma_{xv}(m)] \quad (\text{A.26})$$

As in the case of a continuous-time random process, a discrete-time random process has infinite energy but a finite average power and is given as

$$E(X_n^2) = \gamma_{xx}(0) \quad (\text{A.27})$$

By use of the Wiener-Khinchine theorem, we obtain the power density spectrum of the discrete-time random process by computing the Fourier transform of the autocorrelation sequence $\gamma_{xx}(m)$, that is,

$$\Gamma_{xx}(f) = \sum_{m=-\infty}^{\infty} \gamma_{xx}(m) e^{-j2\pi f m} \quad (\text{A.28})$$

The inverse transform relationship is

$$\gamma_{xx}(m) = \int_{-1/2}^{1/2} \Gamma_{xx}(f) e^{j2\pi f m} df \quad (\text{A.29})$$

We observe that the average power is

$$\gamma_{xx}(0) = \int_{-1/2}^{1/2} \Gamma_{xx}(f) df \quad (\text{A.30})$$

so that $\Gamma_{xx}(f)$ is the distribution of power as a function of frequency, that is, $\Gamma_{xx}(f)$ is the power density spectrum of the random process $X(n)$. The properties we have stated for $\Gamma_{xx}(F)$ also hold for $\Gamma_{xx}(f)$.

Time Averages for a Discrete-Time Random Process

Although we have characterized a random process in terms of statistical averages, such as the mean and the autocorrelation sequence, in practice, we usually have available a single realization of the random process. Let us consider the problem of obtaining the averages of the random process from a single realization. To accomplish this, the random process must be *ergodic*.

By definition, a random process $X(n)$ is ergodic if, with probability 1, all the statistical averages can be determined from a single sample function of the process. In effect, the random process is ergodic if time averages obtained from a single realization are equal to the statistical (ensemble) averages. Under this condition we can attempt to estimate the ensemble averages using time averages from a single realization.

To illustrate this point, let us consider the estimation of the mean and the autocorrelation of the random process from a single realization $x(n)$. Since we are interested only in these two moments, we define ergodicity with respect to these parameters. For additional details on the requirements for mean ergodicity and autocorrelation ergodicity which are given below, the reader is referred to the book of Papoulis (1984).

Mean-Ergodic Process

Given a stationary random process $X(n)$ with mean

$$m_x = E(X_n)$$

let us form the *time average*

$$\hat{m}_x = \frac{1}{2N+1} \sum_{n=-N}^N x(n) \quad (\text{A.31})$$

In general, we view $\hat{m}_x$ in (A.31) as an estimate of the statistical mean whose value will vary with the different realizations of the random process. Hence $\hat{m}_x$ is a random variable with a PDF $p(\hat{m}_x)$. Let us compute the expected value of $\hat{m}_x$ over all possible realizations of $X(n)$. Since the summation and the expectation are linear operations we can interchange them, so that

$$E(\hat{m}_x) = \frac{1}{2N+1} \sum_{n=-N}^N E[x(n)] = \frac{1}{2N+1} \sum_{n=-N}^N m_x = m_x \quad (\text{A.32})$$

Since the mean value of the estimate is equal to the statistical mean, we say that the estimate $\hat{m}_x$ is unbiased.

Next, we compute the variance of $\hat{m}_x$. We have

$$\text{var}(\hat{m}_x) = E(|\hat{m}_x|^2) - |m_x|^2$$

But

$$\begin{aligned} E(|\hat{m}_x|^2) &= \frac{1}{(2N+1)^2} \sum_{n=-N}^N \sum_{k=-N}^N E[x^*(n)x(k)] \\ &= \frac{1}{(2N+1)^2} \sum_{n=-N}^N \sum_{k=-N}^N \gamma_{xx}(k-n) \\ &= \frac{1}{2N+1} \sum_{m=-2N}^{2N} \left(1 - \frac{|m|}{2N+1}\right) \gamma_{xx}(m) \end{aligned}$$

Therefore,

$$\begin{aligned} \text{var}(\hat{m}_x) &= \frac{1}{2N+1} \sum_{m=-2N}^{2N} \left(1 - \frac{|m|}{2N+1}\right) \gamma_{xx} - |m_x|^2 \\ &= \frac{1}{2N+1} \sum_{m=-2N}^{2N} \left(1 - \frac{|m|}{2N+1}\right) c_{xx}(m) \end{aligned} \quad (\text{A.33})$$

If $\text{var}(m_x) \rightarrow 0$ as $N \rightarrow \infty$, the estimate converges with probability 1 to the statistical mean m_x . Therefore, the process $X(n)$ is mean ergodic if

$$\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{m=-2N}^{2N} \left(1 - \frac{|m|}{2N+1}\right) c_{xx}(m) = 0 \quad (\text{A.34})$$

Under this condition, the estimate $\hat{m}_x$ in the limit as $N \rightarrow \infty$ becomes equal to the statistical mean, that is,

$$m_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n) \quad (\text{A.35})$$

Thus the time-average mean, in the limit as $N \rightarrow \infty$, is equal to the ensemble mean.

A sufficient condition for (A.34) to hold is if

$$\sum_{m=-\infty}^{\infty} |c_{xx}(m)| < \infty \quad (\text{A.36})$$

which implies that $c_{xx}(m) \rightarrow 0$ as $m \rightarrow \infty$. This condition holds for most zero-mean processes encountered in the physical world.

Correlation-Ergodic Processes

Now, let us consider the estimate of the autocorrelation $\gamma_{xx}(m)$ from a single realization of the process. Following our previous notation, we denote the estimate (for a complex-valued signal, in general) as

$$r_{xx}(m) = \frac{1}{2N+1} \sum_{n=-N}^N x^*(n)x(n+m) \quad (\text{A.37})$$

Again, we regard $r_{xx}(m)$ as a random variable for any given lag m , since it is a function of the particular realization. The expected value (mean value over all realizations) is

$$\begin{aligned} E[r_{xx}(m)] &= \frac{1}{2N+1} \sum_{n=-N}^N E[x^*(n)x(n+m)] \\ &= \frac{1}{2N+1} \sum_{n=-N}^N \gamma_{xx}(m) = \gamma_{xx}(m) \end{aligned} \quad (\text{A.38})$$

Therefore, the expected value of the time-average autocorrelation is equal to the statistical average. Hence we have an unbiased estimate of $\gamma_{xx}(m)$.

To determine the variance of the estimate $r_{xx}(m)$, we compute the expected value of $|r_{xx}(m)|^2$ and subtract the square of the mean value. Thus

$$\text{var}[r_{xx}(m)] = E[|r_{xx}(m)|^2] - |\gamma_{xx}(m)|^2 \quad (\text{A.39})$$

But

$$E[|r_{xx}(m)|^2] = \frac{1}{(2N+1)^2} \sum_{n=-N}^N \sum_{k=-N}^N E[x^*(n)x(n+m)x(k)x^*(k+m)] \quad (\text{A.40})$$

The expected value of the term $x^*(n)x(n+m)x(k)x^*(k+m)$ is just the autocorrelation sequence of a random process defined as

$$v_m(n) = x^*(n)x(n+m)$$

Hence (A.40) may be expressed as

$$\begin{aligned} E[|r_{xx}(m)|^2] &= \frac{1}{(2N+1)^2} \sum_{n=-N}^N \sum_{k=-N}^N \gamma_{vv}^{(m)}(n-k) \\ &= \frac{1}{2N+1} \sum_{n=-2N}^{2N} \left(1 - \frac{|n|}{2N+1}\right) \gamma_{vv}^{(m)}(n) \end{aligned} \quad (\text{A.41})$$

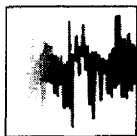
and the variance is

$$\text{var}[r_{xx}(m)] = \frac{1}{2N+1} \sum_{n=-2N}^{2N} \left(1 - \frac{|n|}{2N+1}\right) \gamma_{vv}^{(m)}(n) - |\gamma_{xx}(m)|^2 \quad (\text{A.42})$$

If $\text{var}[r_{xx}(m)] \rightarrow 0$ as $N \rightarrow \infty$, the estimate $r_{xx}(m)$ converges with probability 1 to the statistical autocorrelation $\gamma_{xx}(m)$. Under these conditions, the process is correlation ergodic and the time-average correlation is identical to the statistical average, that is,

$$\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x^*(n)x(n+m) = \gamma_{xx}(m) \quad (\text{A.43})$$

In our treatment of random signals, we assume that the random processes are mean ergodic and correlation ergodic, so that we can deal with time averages of the mean and the autocorrelation obtained from a single realization of the process.



Appendix B

Random Number Generators

In some of the examples given in the text, random numbers are generated to simulate the effect of noise on signals and to illustrate how the method of correlation can be used to detect the presence of a signal buried in noise. In the case of periodic signals, the correlation technique also allowed us to estimate the period of the signal.

In practice, random number generators are often used to simulate the effect of noiselike signals and other random phenomena encountered in the physical world. Such noise is present in electronic devices and systems and usually limits our ability to communicate over large distances and to be able to detect relatively weak signals. By generating such noise on a computer, we are able to study its effects through simulation of communication systems, radar detection systems, and the like and to assess the performance of such systems in the presence of noise.

Most computer software libraries include a uniform random number generator. Such a random number generator generates a number between zero and 1 with equal probability. We call the output of the random number generator a random variable. If A denotes such a random variable, its range is the interval $0 \leq A \leq 1$.

We know that the numerical output of a digital computer has limited precision, and as a consequence, it is impossible to represent the continuum of numbers in the interval $0 \leq A \leq 1$. However, we can assume that our computer represents each output by a large number of bits in either fixed point or floating point. Consequently, for all practical purposes, the number of outputs in the interval $0 \leq A \leq 1$ is sufficiently large, so that we are justified in assuming that any value in the interval is a possible output from the generator.

The uniform probability density function for the random variable A , denoted as $p(A)$, is illustrated in Fig. B.1a. We note that the average value or mean value of A , denoted as m_A , is $m_A = \frac{1}{2}$. The integral of the probability density function, which represents the area under $p(A)$, is called the probability distribution function

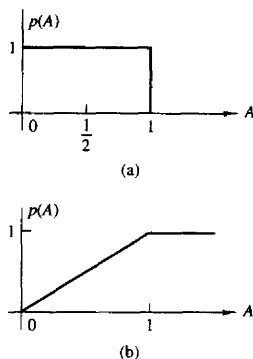


Figure B.1

of the random variable A and is defined as

$$F(A) = \int_{-\infty}^A p(x) dx \quad (\text{B.1})$$

For any random variable, this area must always be unity, which is the maximum value that can be achieved by a distribution function. Hence

$$F(1) = \int_{-\infty}^1 p(x) dx = 1 \quad (\text{B.2})$$

and the range of $F(A)$ is $0 \leq F(A) \leq 1$ for $0 \leq A \leq 1$.

If we wish to generate uniformly distributed noise in an interval $(b, b+1)$ it can simply be accomplished by using the output A of the random number generator and shifting it by an amount b . Thus a new random variable B can be defined as

$$B = A + b \quad (\text{B.3})$$

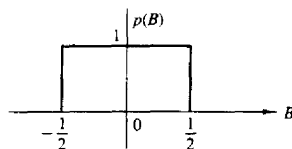
which now has a mean value $m_B = b + \frac{1}{2}$. For example, if $b = -\frac{1}{2}$, the random variable B is uniformly distributed in the interval $(-\frac{1}{2}, \frac{1}{2})$, as shown in Fig. B.2a. Its probability distribution function $F(B)$ is shown in Fig. B.2b.

A uniformly distributed random variable in the range $(0, 1)$ can be used to generate random variables with other probability distribution functions. For example, suppose that we wish to generate a random variable C with probability distribution function $F(C)$, as illustrated in Fig. B.3. Since the range of $F(C)$ is the interval $(0, 1)$, we begin by generating a uniformly distributed random variable A in the range $(0, 1)$. If we set

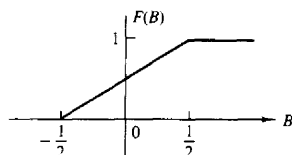
$$F(C) = A \quad (\text{B.4})$$

then

$$C = F^{-1}(A) \quad (\text{B.5})$$



(a)



(b)

Figure B.2

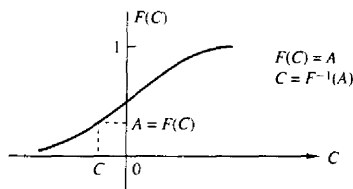


Figure B.3

Thus we solve (B.4) for C and the solution in (B.5) provides the value of C for which $F(C) = A$. By this means we obtain a new random variable C with probability distribution $F(C)$. This inverse mapping from A to C is illustrated in Fig. B.3.

Example B.1

Generate a random variable C that has the linear probability density function shown in Fig. B.4a, that is,

$$p(C) = \begin{cases} \frac{C}{2}, & 0 \leq C \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Solution This random variable has a probability distribution function

$$F(C) = \begin{cases} 0, & C < 0 \\ \frac{1}{4}C^2, & 0 \leq C \leq 2 \\ 1, & C > 2 \end{cases}$$

which is illustrated in Fig. B.4b. We generate a uniformly distributed random variable A and set $F(C) = A$. Hence

$$F(C) = \frac{1}{4}C^2 = A$$

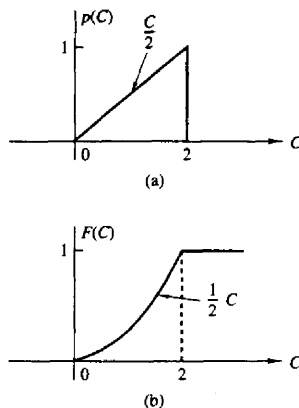


Figure B.4

Upon solving for C , we obtain

$$C = 2\sqrt{A}$$

Thus we generate a random variable C with probability function $F(C)$, as shown in Fig. B.4b.

In Example B.1 the inverse mapping $C = F^{-1}(A)$ was simple. In some cases it is not. This problem arises in trying to generate random numbers that have a normal distribution function.

Noise encountered in physical systems is often characterized by the normal or Gaussian probability distribution, which is illustrated in Fig. B.5. The probability

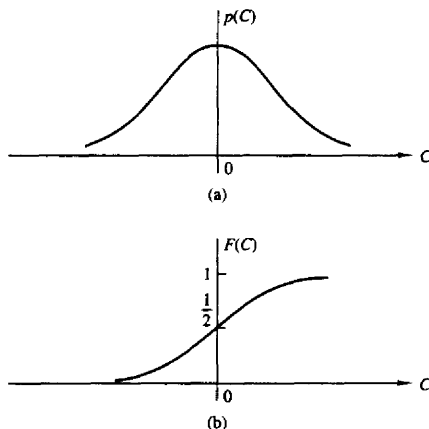


Figure B.5

density function is given by

$$p(C) = \frac{1}{\sqrt{2\pi}\sigma} e^{-C^2/2\sigma^2} \quad -\infty < C < \infty \quad (\text{B.6})$$

where σ^2 is the variance of C , which is a measure of the spread of the probability density function $p(C)$. The probability distribution function $F(C)$ is the area under $p(C)$ over the range $(-\infty, C)$. Thus

$$F(C) = \int_{-\infty}^C p(x) dx \quad (\text{B.7})$$

Unfortunately, the integral in (B.7) cannot be expressed in terms of simple functions. Consequently, the inverse mapping is difficult to achieve.

A way has been found to circumvent this problem. From probability theory it is known that a (Rayleigh distributed) random variable R , with probability distribution function

$$F(R) = \begin{cases} 0, & R < 0 \\ 1 - e^{-R^2/2\sigma^2}, & R \geq 0 \end{cases} \quad (\text{B.8})$$

is related to a pair of Gaussian random variables C and D , through the transformation

$$C = R \cos \Theta \quad (\text{B.9})$$

$$D = R \sin \Theta \quad (\text{B.10})$$

where Θ is a uniformly distributed variable in the interval $(0, 2\pi)$. The parameter σ^2 is the variance of C and D . Since (B.8) is easily inverted, we have

$$F(R) = 1 - e^{-R^2/2\sigma^2} = A$$

and hence

$$R = \sqrt{2\sigma^2 \ln[1/(1 - A)]} \quad (\text{B.11})$$

where A is a uniformly distributed random variable in the interval $(0, 1)$. Now if we generate a second uniformly distributed random variable B and define

$$\Theta = 2\pi B \quad (\text{B.12})$$

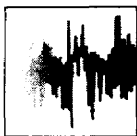
then from (B.9) and (B.10), we obtain two statistically independent Gaussian distributed random variables C and D .

The method described above is often used in practice to generate Gaussian distributed random variables. As shown in Fig. B.5, these random variables have a mean value of zero and a variance σ^2 . If a nonzero mean Gaussian random variable is desired, then C and D can be translated by the addition of the mean value.

A subroutine implementing this method for generating Gaussian distributed random variables is given in Fig. B.6.


```
C      SUBROUTINE GAUSS CONVERTS A UNIFORM RANDOM
C      SEQUENCE XIN IN [0,1] TO A GAUSSIAN RANDOM
C      SEQUENCE WITH G(0,SIGMA**2)
C      PARAMETERS      :
C          XIN          :UNIFORM IN [0,1] RANDOM NUMBER
C          B             :UNIFORM IN [0,1] RANDOM NUMBER
C          SIGMA         :STANDARD DEVIATION OF THE GAUSSIAN
C          YOUT          :OUTPUT FROM THE GENERATOR
C
C      SUBROUTINE GAUSS 9XIN,B,SIGMA,YOUT)
C      PI=4.0*ATAN (1.0)
C      B=2.0*PI*B
C      R=SQRT (2.0*(SIGMA**2)*ALOG(1.0/(1.0-XIN)))
C      YOUT=R*COS(B)
C      RETURN
C      END
C
C      NOTE: TO USE THE ABOVE SUBROUTINE FOR A
C      GAUSSIAN RANDOM NUMBER GENERATOR
C      YOU MUST PROVIDE AS INPUT TWO UNIFORM RANDOM NUMBERS
C      XIN AND B
C      XIN AND B MUST BE STATISTICALLY INDEPENDENT
C
```

Figure B.6 Subroutine for generating Gaussian random variables



Appendix

C

Tables of Transition Coefficients for the Design of Linear-Phase FIR Filters

In Section 8.2.3 we described a design method for linear-phase FIR filters that involved the specification of $H_r(\omega)$ at a set of equally spaced frequencies $\omega_k = 2\pi(k + \alpha)/M$, where $\alpha = 0$ or $\alpha = \frac{1}{2}$, $k = 0, 1, \dots, (M-1)/2$ for M odd and $k = 0, 1, 2, \dots, (M/2) - 1$ for M even, where M is the length of the filter. Within the passband of the filter, we select $H_r(\omega_k) = 1$, and in the stopband, $H_r(\omega_k) = 0$. For frequencies in the transition band, the values of $H_r(\omega_k)$ are optimized to minimize the maximum sidelobe in the stopband. This is called a *minimax optimization criterion*.

The optimization of the values of $H_r(\omega)$ in the transition band has been performed by Rabiner et al. (1970) and tables of transition values have been provided in the published paper. A selected number of the tables for lowpass FIR filters are included in this appendix.

Four tables are given. Table C.1 lists the transition coefficients for the case $\alpha = 0$ and one coefficient in the transition band for both M odd and M even. Table C.2 lists the transition coefficients for the case $\alpha = 0$, and two coefficients in the transition band for M odd and M even. Table C.3 lists the transition coefficients for the case $\alpha = \frac{1}{2}$, M even and one coefficient in the transition band. Finally, Table C.4 lists the transition coefficients for the case $\alpha = \frac{1}{2}$, M even, and two coefficients in the transition band. The tables also include the level of the maximum sidelobe and a bandwidth parameter, denoted as BW.

To use the tables, we begin with a set of specifications, including (1) the bandwidth of the filter, which can be defined as $(2\pi/M)(BW + \alpha)$, where BW is the number of consecutive frequencies at which $H(\omega_k) = 1$, (2) the width of the transition region, which is roughly $2\pi/M$ times the number of transition coefficients, and (3) the maximum tolerable sidelobe in the stopband. The length of the filter can be selected from the tables to satisfy the specifications.

TABLE C.1 TRANSITION COEFFICIENTS FOR $\alpha = 0$

<i>M</i> Odd			<i>M</i> Even		
BW	Minimax	T_1	BW	Minimax	T_1
<i>M</i> = 15			<i>M</i> = 16		
1	-42.30932283	0.43378296	1	-39.75363827	0.42631836
2	-41.26299286	0.41793823	2	-37.61346340	0.40397949
3	-41.25333786	0.41047636	3	-36.57721567	0.39454346
4	-41.94907713	0.40405884	4	-35.87249756	0.38916626
5	-44.37124538	0.39268189	5	-35.31695461	0.38840332
6	-56.01416588	0.35766525	6	-35.51951933	0.40155639
<i>M</i> = 33			<i>M</i> = 32		
1	-43.03163004	0.42994995	1	-42.24728918	0.42856445
2	-42.42527962	0.41042481	2	-41.29370594	0.40773926
3	-42.40898275	0.40141601	3	-41.03810358	0.39662476
4	-42.45948601	0.39641724	4	-40.93496323	0.38925171
6	-42.52403450	0.39161377	5	-40.85183477	0.37897949
8	-42.44085121	0.39039917	8	-40.75032616	0.36990356
10	-42.11079407	0.39192505	10	-40.54562140	0.35928955
12	-41.92705250	0.39420166	12	-39.93450451	0.34487915
14	-44.69430351	0.38552246	14	-38.91993237	0.34407349
15	-56.18293285	0.35360718			
<i>M</i> = 65			<i>M</i> = 64		
1	-43.16935968	0.42919312	1	-42.96059322	0.42882080
2	-42.61945581	0.40903320	2	-42.30815172	0.40830689
3	-42.70906305	0.39920654	3	-42.32423735	0.39807129
4	-42.86997318	0.39335937	4	-42.43565893	0.39177246
5	-43.01999664	0.38950806	5	-42.55461407	0.38742065
6	-43.14578819	0.38679809	6	-42.66526604	0.38416748
10	-43.44808340	0.38129272	10	-43.01104736	0.37609863
14	-43.54684496	0.37946167	14	-43.28309965	0.37089233
18	-43.48173618	0.37955322	18	-43.56508827	0.36605225
22	-43.19538212	0.38162842	22	-43.96245098	0.35977783
26	-42.44725609	0.38746948	26	-44.60516977	0.34813232
30	-44.76228619	0.38417358	30	-43.81448936	0.29973144
31	-59.21673775	0.35282745			
<i>M</i> = 125			<i>M</i> = 128		
1	-43.20501566	0.42899170	1	-43.15302420	0.42889404
2	-42.66971111	0.40867310	2	-42.59092569	0.40847778
3	-42.77438974	0.39868774	3	-42.67634487	0.39838257
4	-42.95051050	0.39268189	4	-42.84038544	0.39226685
6	-43.25854683	0.38579101	5	-42.99805641	0.38812256
8	-43.47917461	0.38195801	7	-43.25537014	0.38281250
10	-43.63750410	0.37954102	10	-43.52547789	0.3782638
18	-43.95589399	0.37518311	18	-43.93180990	0.37251587
26	-44.05913115	0.37384033	26	-44.18097305	0.36941528
34	-44.05672455	0.37371826	34	-44.40153408	0.36686401
42	-43.94708776	0.37470093	42	-44.67161417	0.36394653
50	-43.58473492	0.37797851	50	-45.17186594	0.35902100
58	-42.14925432	0.39086304	58	-46.92415667	0.34273681
59	-42.60623264	0.39063110	62	-49.46298973	0.28751221
60	-44.78062010	0.38383713			
61	-56.22547865	0.35263062			

Source: Rabiner et al. (1970); © 1970 IEEE; reprinted with permission.

TABLE C.2 TRANSITION COEFFICIENTS FOR $\alpha = 0$

<i>M</i> Odd				<i>M</i> Even			
BW	Minimax	T_1	T_2	BW	Minimax	T_1	T_2
<i>M</i> = 15				<i>M</i> = 16			
1	-70.60540585	0.09500122	0.58995418	1	-65.27693653	0.10703125	0.60559357
2	-69.26168156	0.10319824	0.59357118	2	-62.85937929	0.12384644	0.62201631
3	-69.91973495	0.10083618	0.58594327	3	-62.96594906	0.12827148	0.62855407
4	-75.51172256	0.08407953	0.55715312	4	-66.03942485	0.12130127	0.61952704
5	-103.45078300	0.05180206	0.49917424	5	-71.73997498	0.11066284	0.60979204
<i>M</i> = 33				<i>M</i> = 32			
1	-70.60967541	0.09497070	0.58985167	1	-67.37020397	0.09610596	0.59045212
2	-68.16726971	0.10585937	0.59743846	2	-63.93104696	0.11263428	0.60560235
3	-67.13149548	0.10937500	0.59911696	3	-62.49787903	0.11931763	0.61192546
5	-66.53917217	0.10965576	0.59674101	5	-61.28204536	0.12541504	0.61824023
7	-67.23387909	0.10902100	0.59417456	7	-60.82049131	0.12907715	0.62307031
9	-67.85412312	0.10502930	0.58771575	9	-59.74928167	0.12068481	0.60685586
11	-69.08597469	0.10219727	0.58216391	11	-62.48683357	0.13004150	0.62821502
13	-75.86953640	0.08137207	0.54712777	13	-70.64571857	0.11017914	0.60670943
14	-104.04059029	0.05029373	0.49149549				
<i>M</i> = 65				<i>M</i> = 64			
1	-70.66014957	0.09472656	0.58945943	1	-70.26372528	0.09376831	0.58789222
2	-68.89622307	0.10404663	0.59476127	2	-67.20729542	0.10411987	0.59421778
3	-67.90234470	0.10720215	0.59577449	3	-65.80684280	0.10850220	0.59666158
4	-67.24003792	0.10726929	0.59415763	4	-64.95227051	0.11038818	0.59730067
5	-66.86065960	0.10689087	0.59253047	5	-64.42742348	0.11113281	0.59698496
9	-66.27561188	0.10548706	0.58845983	9	-63.41714096	0.10936890	0.59088884
13	-65.96417046	0.10466309	0.58660485	13	-62.72142410	0.10828857	0.58738641
17	-66.16404629	0.10649414	0.58862042	17	-62.37051868	0.11031494	0.58968142
21	-66.76456833	0.10701904	0.58894575	21	-62.04848146	0.11254273	0.59249461
25	-68.13407993	0.10327148	0.58320831	25	-61.88074064	0.11994629	0.60564501
29	-75.98313046	0.08069458	0.54500379	29	-70.05681992	0.10717773	0.59842159
30	-104.92083740	0.04978485	0.48965181				
<i>M</i> = 125				<i>M</i> = 128			
1	-70.68010235	0.09464722	0.58933268	1	-70.58992958	0.09445190	0.58900996
2	-68.94157696	0.10390015	0.59450024	2	-68.62421608	0.10349731	0.59379058
3	-68.19352627	0.10682373	0.59508549	3	-67.66701698	0.10701294	0.59506081
5	-67.34261131	0.10668945	0.59187505	4	-66.95196629	0.10685425	0.59289826
7	-67.09767151	0.10587158	0.59821869	6	-66.32718945	0.10596924	0.58953845
9	-67.05801296	0.10523682	0.58738706	9	-66.01315498	0.10471191	0.58593906
17	-67.17504501	0.10372925	0.58358265	17	-65.89422417	0.10288086	0.58097354
25	-67.22918987	0.10316772	0.58224835	25	-65.92644215	0.10182495	0.57812308
33	-67.11609936	0.10303955	0.58198956	33	-65.95577812	0.10096436	0.57576437
41	-66.71271324	0.10313721	0.58245499	41	-65.97698021	0.10094604	0.57451694
49	-66.62364197	0.10561523	0.58629534	49	-65.67919827	0.09865112	0.56927420
57	-69.28378487	0.10061646	0.57812192	57	-64.61514568	0.09845581	0.56604486
58	-70.35782337	0.09663696	0.57121235	61	-71.76589394	0.10496826	0.59452277
59	-75.94707718	0.08054886	0.54451285				
60	-104.09012318	0.04991760	0.48963264				

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TABLE C.3 TRANSITION
COEFFICIENTS FOR $\alpha = \frac{1}{2}$

BW	Minimax	T_1
<i>M</i> = 16		
1	-51.60668707	0.26674805
2	-47.48000240	0.32149048
3	-45.19746828	0.34810181
4	-44.32862616	0.36308594
5	-45.68347692	0.36661987
6	-56.63700199	0.34327393
<i>M</i> = 32		
1	-52.64991188	0.26073609
2	-49.39390278	0.30878296
3	-47.72596645	0.32984619
4	-46.68811989	0.34217529
6	-45.33436489	0.35704956
8	-44.30730963	0.36750488
10	-43.11168003	0.37810669
12	-42.97900438	0.38465576
14	-56.32780266	0.35030518
<i>M</i> = 64		
1	-52.90375662	0.25923462
2	-49.74046421	0.30603638
3	-48.38088989	0.32510986
4	-47.47863007	0.33595581
5	-46.88655186	0.34287720
6	-46.46230555	0.34774170
10	-45.46141434	0.35859375
14	-44.85988188	0.36470337
18	-44.34302616	0.36983643
22	-43.69835377	0.37586059
26	-42.45641375	0.38624268
30	-56.25024033	0.35200195
<i>M</i> = 128		
1	-52.96778202	0.25885620
2	-49.82771969	0.30534668
3	-48.51341629	0.32404785
4	-47.67455149	0.33443604
5	-47.11462021	0.34100952
7	-46.43420267	0.34880371
10	-45.88529110	0.35493774
18	-45.21660566	0.36182251
26	-44.87959814	0.36521607
34	-44.61497784	0.36784058
42	-44.32706451	0.37066040
50	-43.87646437	0.37500000
58	-42.30969715	0.38807373
62	-56.23294735	0.35241699

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TABLE C.4 TRANSITION COEFFICIENTS FOR
 $\alpha = \frac{1}{2}$

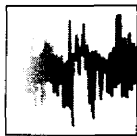
BW	Minimax	T_1	T_2
$M = 16$			
1	-77.26126766	0.05309448	0.41784180
2	-73.81026745	0.07175293	0.49369211
3	-73.02352142	0.07862549	0.51966134
4	-77.95156193	0.07042847	0.51158076
5	-105.23953247	0.04587402	0.46967784
$M = 32$			
1	-80.49464130	0.04725342	0.40357383
2	-73.92513466	0.07094727	0.49129255
3	-72.40863037	0.08012695	0.52153983
5	-70.95047379	0.08935547	0.54805908
7	-70.22383976	0.09403687	0.56031410
9	-69.94402790	0.09628906	0.56637987
11	-70.82423878	0.09323731	0.56226952
13	-104.85642624	0.04882812	0.48479068
$M = 64$			
1	-80.80974960	0.04658203	0.40168723
2	-75.11772251	0.06759644	0.48390015
3	-72.66662025	0.07886963	0.51850058
4	-71.85610867	0.08393555	0.53379876
5	-71.34401417	0.08721924	0.54311474
9	-70.32861614	0.09371948	0.56020256
13	-69.34809303	0.09761963	0.56903714
17	-68.06440258	0.10051880	0.57543691
21	-67.99149132	0.10289307	0.58007699
25	-69.32065105	0.10068359	0.57729656
29	-105.72862339	0.04923706	0.48767025
$M = 128$			
1	-80.89347839	0.04639893	0.40117195
2	-77.22580583	0.06295776	0.47399521
3	-73.43786240	0.07648926	0.51361278
4	-71.93675232	0.08345947	0.53266251
6	-71.10850430	0.08880615	0.54769675
9	-70.53600121	0.09255371	0.55752959
17	-69.95890045	0.09628906	0.56676912
25	-69.29977322	0.09834595	0.57137301
33	-68.75139713	0.10077515	0.57594641
41	-67.89687920	0.10183716	0.57863142
49	-66.76120186	0.10264282	0.58123560
57	-69.21525860	0.10157471	0.57946395
61	-104.57432938	0.04970703	0.48900685

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As an illustration, the filter design for which $M = 15$ and

$$H_r\left(\frac{2\pi k}{M}\right) = \begin{cases} 1, & k = 0, 1, 2, 3 \\ T_1, & k = 4 \\ 0, & k = 5, 6, 7 \end{cases}$$

corresponds to $\alpha = 0$, $BW = 4$, since $H_r(\omega_k) = 1$ at the four consecutive frequencies $\omega_k = 2\pi k/15$, $k = 0, 1, 2, 3$, and the transition coefficient is T_1 at the frequency $\omega_k = 8\pi/15$. The value given in Table C.1 for $M = 15$ and $BW = 4$ is $T_1 = 0.40405884$. The maximum sidelobe is at -41.9 dB, according to Table C.1.



Appendix D

List of MATLAB Functions

In this Appendix, we list several MATLAB functions that the student can use to solve some of the problems numerically. The list includes the most relevant MATLAB functions for each of the chapters, but it is not exhaustive. However, this list is cumulative in the sense that once a function is listed in any chapter, it is not repeated in subsequent chapters. These MATLAB functions are obtained from two sources: (1) the student version of MATLAB and (2) the book entitled *Digital Signal Processing Using MATLAB*. (PWS Kent 1996), by V.K. Ingle and J.G. Proakis.

Our primary objective in listing these MATLAB functions is to inform the student who is not familiar with MATLAB of the existence of these functions and to encourage the student to use them in the solution of some of the homework problems.

CHAPTER 1

sin(x), cos(x), tan(x)	trigonometric functions
abs(x)	absolute values of a vector x with real or complex components.
real(x)	takes the real part of each components of the vector x .
imag(x)	takes the imaginary part of each component of the vector x .
conj(x)	complex-conjugate of each component of x .
exp(z)	$e^x(\cos y + j \sin y)$, where $z = x + jy$.
sum(x)	sum of the (real or complex) components of the vector x .
prod(x)	product of the (real or complex) components of the vector x .
angle(x)	computes the phase angles of each component of the vector x .
log(x)	computes the natural logarithm of each of the elements of x .

log10(x)	computes the logarithm to the base 10 of the elements of x .
sqrt(x)	computes the square root of the elements of x .

CHAPTER 2

conv(x, h)	convolution of the two (vector) sequences x and h .
fliplr(x)	folds the (vector) sequence x .
filter(b, a, x)	solves the difference equation with coefficients

$$a = [a_0, a_1, \dots, a_n]$$

$$b = [b_0, b_1, \dots, b_m]$$

$$x = \text{input sequence}$$

filter(b, 1, x)	implements an FIR filter with input x and coefficients b .
rand(1, N)	generates a length N random sequence that is uniform in the interval (0, 1).
randn(1, N)	generates a length N sequence of Gaussian random variables with zero mean and unit variance.
xcorr(x, y)	computes the crosscorrelation of the two sequences x and y .
xcorr(x)	computes the autocorrelation of the sequence x .

CHAPTER 3

roots(a)	computes the roots of the polynomial with coefficients
residuez(b, a)	computes the residues in a partial fraction expansion, where

$$b = \text{coefficients of numerator polynomial} \\ [b_0, b_1, \dots, b_M]$$

$$a = \text{coefficients of denominator polynomial} \\ [a_0, a_1, \dots, a_N]$$

deconv(b, a)	computes the result of dividing b by a in a polynomial part p and a remainder r .
poly(r)	computes the coefficients of the polynomial p with roots r .
pzplotz(b, a)	plots the poles and zeros in the z -plane given the coefficient vectors b and a .

filter(b, a, x, xic) implements the filter given by a difference equation with coefficient vector b and a , input x and initial conditions x_{ic} .

CHAPTER 4

freqz(b, a, N) computes an N -point complex frequency response vector and an N -point frequency vector ω , uniform over the interval $0 < \omega < \pi$, for filter with coefficient vector b and a .

freqz(b, a, N, 'whole') same computation as **freqz(b, a, N)**, except that the frequency range is $0 < \omega < 2\pi$.

freqz(b, a, ω) computes the frequency response of the system at the frequencies specified by the vector ω .

grpdelay(b, a, N) computes the group delay of the filter with numerator polynomial having coefficients b and denominator polynomial with coefficients a , at N points over the interval $(0, \pi)$.

grpdelay(b, a, N, 'whole') same as above, except that the frequency range is $0 < \omega < 2\pi$.

CHAPTER 5

dfs(x, N) computes the discrete fourier series (DFS) coefficient array for the periodic signal sequence x with period N .

idfs(y, N) computes the signal sequence from the DFS coefficient array y .

rem(n, N) determines the remainder after dividing n by N .

mod(n, N) computes $n \bmod N$.

dft(x, N) computes the N -point DFT of the data sequence x .

idft(X, N) computes the N -point inverse DFT of X .

ovrlpsav(x, h, N) implements the overlap-save method to perform block convolution where N is the block length.

CHAPTER 6

fft(x, N) implements a radix-2, N -point FFT algorithm.

ifft(X, N) implements a radix-2, N -point inverse FFT algorithm.

fftshift(x) rearranges the outputs of **fft** so that the zero frequency component is the center of the spectrum.

CHAPTER 7

dir2cas(b, a) converts a direct form structure to the cascade form.

cas2dir (b_0, B, A)	converts a cascade structure to the direct form structure.
dir2par (b, a)	converts direct form to the parallel form structure.
par2dir (C_0B, A)	converts a parallel form to the direct form structure.
dir2latc (b)	converts a FIR direct form structure to an all-zero lattice structure.
latc2dir (K)	converts an all-zero lattice structure to the direct form structure.
dir2ladr (b, a)	converts direct form IIR structure to pole-zero lattice-ladder structure.
ladr2dir (K, C)	converts a lattice-ladder structure to the direct form IIR structure.
casfilt (b_0, B, A, x)	implements the cascade form IIR and FIR realization of a filter with input sequence x .
parfilt (C, B, A, x)	implements the parallel form IIR realization of a filter with input sequence x .
latcfilt (K, x)	implements the FIR lattice filter realization with input sequence x .
ladrfilt (K, C, x)	implements the lattice-ladder realization of a filter with input sequence x .
round (x)	rounds the components of the vector x to the nearest integer
fix (x)	rounds (truncates) the component of the vector x to the nearest integer toward zero.
sign (x)	each component of x is set to +1 if it is positive and -1 if it is negative.
ss2tf (A, B, C, D, iu)	computes the transfer function $H(x)$ of a system given the state-space description of the form $\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$ from the iu th input.
ss2zp (A, D, C, D, iu)	computes the transfer function $H(s)$ and expresses it in factored form, thus, giving the poles and zeros of $H(s)$.

CHAPTER 8

boxcar (M)	generates an M -point rectangular window.
bartlett (M)	generates an M -point Bartlett window.
hanning (M)	generates an M -point Hanning window.

hamming(M)	generates an M -point Hamming window.
blackman(M)	generates an M -point Blackman window.
kaiser(M)	generates an M -point kaiser window.
buttap(N)	provides the coefficients of an analog lowpass Butterworth filter of order N , with normalized frequency, in cascade form.
chebiap(N, R_p)	provides the coefficients of an analog lowpass Chebyshev filter of order N , with normalized frequency and passband ripple R_p , in cascade form.
ellipap(N, R_p, A_s)	provides the coefficients of an analog lowpass elliptic filter of order N , passband ripple R_p , stopband attenuation A_s , with normalized frequency, in cascade form.
freqs(b, a, ω)	computes the frequency response of an analog filter, with ω in rad/sec.
butter(N, ω_n)	designs a digital lowpass Butterworth filter of order N and cutoff frequency ω_n .
cheby1(N, R_p, ω_n)	designs a digital lowpass Chebyshev filter of order N passband ripple R_p , and cutoff frequency ω_n .
cheby2(N, A_s, ω_n)	designs a Type 2 lowpass Chebyshev filter of order N , stopband ripple A_s , and cutoff frequency ω_n .
ellip(N, R_p, A_s, ω_n)	designs a digital lowpass elliptic filter of order N , passband ripple R_p , stopband ripple A_s , and cutoff frequency ω_n .
bilinear(z, p, k, fs)	uses the bilinear transformation to convert an analog filter with zeros z , poles p , and gain k , into a digital filter, with fs being the sample frequency in Hz.
bilinear(num, den, fs)	uses the bilinear transformation to convert an analog filter with numerator polynomial coefficients num , denominator polynomial coefficients den , and sample frequency fs , into a digital filter.
remez(N, f, m)	uses the Remez algorithm to determine the coefficients of an optimum equiripple, linear phase FIR filter of length $N + 1$, from frequency specifications f and gains m for each band.
remez($N, f, m, \text{'ftype'}$)	same description as above with 'ftype' used to specify a Hilbert transform or differentiator.
butter($N, \omega_n, \text{'high'}$)	designs a highpass Butterworth filter of order N and 3-dB cutoff frequency ω_n .
butter($N, \omega_n, \text{'bandpass'}$)	designs a $2N$ -order bandpass Butterworth filter, with 3-dB passband $\omega_1 < \omega < \omega_2$, where $\omega_n = [\omega_1, \omega_2]$.
cheby1($N, R_p, \omega_n, \text{'high'}$)	designs a highpass Chebyshev filter of order N , passband ripple R_p , and cutoff frequency ω_n .

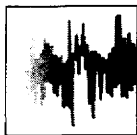
ellip ($N, R_p, A_s, \omega n$)	designs an elliptic bandpass filter of order N , pass-band ripple R_p , stopband attenuation A_s , and cutoff frequencies $\omega n = [\omega 1, \omega 2]$.
lp2bp (num , den , ωo , $B\omega$)	transforms an analog lowpass filter to an analog band-pass filter.
lp2bs (num , den , ωo , $B\omega$)	transforms an analog lowpass filter to an analog band-stop filter.
lp2hp (num , den , ωo)	transforms an analog lowpass filter to an analog high-pass filter.
lp2lp (num , den , ωo)	transforms an analog lowpass filter to an analog low-pass filter with cutoff frequency ωo .
polyfit (x, y, n)	finds a polynomial p such that $p(x)$ fits the data in a vector y in a least squares sense.

CHAPTER 9

spline (x, y, xi)	cubic spline data interpolation.
spline (nt_s, x, t)	uses cubic spline interpolation, where x and nt_s are arrays containing samples $x(n)$ at nt_s , and t is an array that contains a fine grid at which the function is evaluated.

CHAPTER 10

dnsample (x, M)	downsamples the sequence x by the factor M .
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